L2 SERIES

by S. Naz

Complete IT Infrastructure & System Administration Lab Guide

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- Active Directory User Management
- Group Policy Troubleshooting
- DNS Issue Troubleshooting
- DHCP Failure Troubleshooting
- File Server Permission Management
- Print Server Troubleshooting
- Outlook Issues & Solutions
- Office 365 Administration
- OneDrive & SharePoint Collaboration
- Microsoft Teams & Planner
- Power Automate Workflows
- Microsoft Purview DLP Policies
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- SCCM Patch Management
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- WSUS Patch Management

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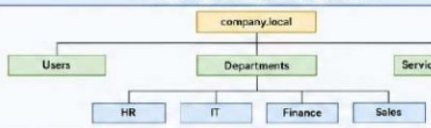
LAB 1: ACTIVE DIRECTORY USER MANAGEMENT

Objective: Create a new user, assign to OU, reset password, unlock account and disable terminated employee account.

LAB ENVIRONMENT

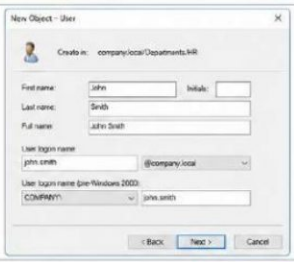
- Windows Server 2022
- Domain: company.local
- Domain Admin Credentials
- Tools: Active Directory Users and Computers (ADUC)

ORGANIZATIONAL UNIT STRUCTURE



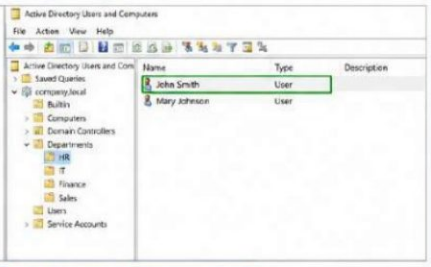
1. CREATE A NEW USER

1. Open Active Directory Users and Computers
2. Right click on the OU (e.g. HR)
3. Click New → User
4. Enter user details:
   - First name: John
   - Last name: Smith
   - User logon name: john.smith
   - User logon name (UPN): john.smith@company.local
5. Click Next
6. Set Password: Welcome123!
7. Uncheck "User must change password at next logon"
8. Click Next → Finish



2. ASSIGN TO DEPARTMENT OU

1. In ADUC, navigate to company.local → Departments → HR
2. The newly created user john.smith will appear under HR OU.
3. This ensures proper organization and GPO application.



3. RESET PASSWORD

1. Right click on the user (john.smith)
2. Click Reset Password...
3. Enter new password: NewPass123!
4. Uncheck "User must change password at next logon"
5. Click OK
6. Click OK to confirm

4. UNLOCK ACCOUNT

1. Right click on the user (if account is locked)
2. Click Unlock Account
3. You will see a confirmation message.

5. DISABLE TERMINATED EMPLOYEE ACCOUNT

1. Right click on the user (terminated employee)
2. Click Disable Account
3. The account will be disabled.
4. The user will no longer be able to log in.

VERIFY ALL TASKS

- ✅ User John Smith created under HR OU
- ✅ Password reset successfully
- ✅ Account unlocked
- ✅ Terminated employee account disabled

SUMMARY OF COMMANDS (Optional PowerShell)

Create User

```
New-ADUser -Name "John Smith" ` `
 -GivenName "John" -Surname "Smith" `
 -SamAccountName "john.smith" ` `
 -UserPrincipalName "john.smith@company.local" `
 -Path "OU=HR,OU=Departments,DC=company,DC=local" `
 -AccountPassword (ConvertTo-SecureString "Welcome123!" `
  -AsPlainText -Force) -Enabled $true
```

Reset Password

```
Set-ADAccountPassword -Identity "john.smith" `
-NewPassword (ConvertTo-SecureString "NewPass123!" `,
 -AsPlainText -Force) -Reset
```

Unlock Account

```
Unlock-ADAccount -Identity "john.smith"
```

Disable Account

```
Disable-ADAccount -Identity "john.smith"
```

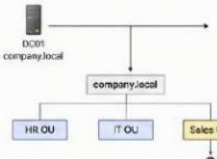
KEY POINTS Always place users in correct OU • Use strong password policy • Unlock accounts instead of creating new one • Disable accounts instead of deleting • Follow least privilege principle

LAB 2: GROUP POLICY TROUBLESHOOTING

Scenario: Users are able to access Control Panel. The requirement is to block access to Control Panel for all users in the Sales OU.

LAB ENVIRONMENT

- Domain: company.local
- Domain Controller: DC01 (Windows Server 2022)
- Client: WIN10 (Windows 10/11)
- Tool: Group Policy Management (GPMC)
- User: sales.user (member of Sales OU)

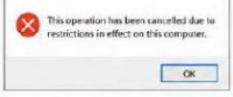


OBJECTIVE

Prevent users in Sales OU from accessing Control Panel.

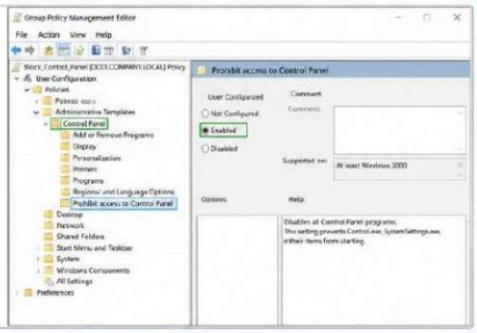
EXPECTED RESULT

Users in Sales OU will get the message:



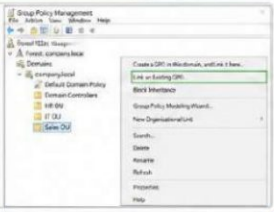
STEP 1: CREATE AND CONFIGURE GPO

1. Open Group Policy Management (gpmc.msc)
2. Right click on Group Policy Objects → New
3. Name it: Block_Control_Panel
4. Right click the GPO → Edit
5. Navigate to:
   User Configuration →
   Policies → Administrative Templates →
   Control Panel → Prohibit access to Control Panel
6. Double click "Prohibit access to Control Panel"
7. Select: Enabled
8. Click Apply → OK



STEP 2: LINK GPO TO SALES OU

1. In GPMC, right click Sales OU
2. Click Link an Existing GPO...
3. Select Block_Control_Panel
4. Click OK
5. The GPO is now linked to Sales OU



STEP 3: FORCE GROUP POLICY UPDATE

1. Log on to client machine WIN10 with sales.user
2. Open Command Prompt as normal user
3. Run the following command:
   `gpupdate /force`
4. You should see:
   "Computer Policy update has completed successfully."
   "User Policy update has completed successfully."

STEP 4: VERIFY RESULT

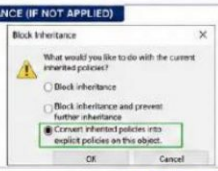
1. Press Windows + R
2. Type control
3. Press Enter
4. You should see the restriction message.

STEP 5: USE GPRESULT TO VERIFY

1. Open Command Prompt
2. Run:
   `gpresult /r`
3. Check the "Applied Group Policy Objects" section. You should see Block_Control_Panel listed.

STEP 6: TROUBLESHOOT INHERITANCE (IF NOT APPLIED)

1. Right click Sales OU → Block Inheritance
2. Select: Convert inherited policies into explicit policies on this object
3. Click OK
4. Run gpupdate /force again



TROUBLESHOOTING TIPS

- Make sure the user is in the Sales OU.
- Ensure the GPO is linked to the correct OU.
- Run gpresult /r to check which GPOs are applied.
- Check for GPO conflicts (Restricted Groups, Preferences, etc.).
- Ensure no other GPO is overriding this setting.
- Allow some time or run gpupdate /force.

VERIFICATION CHECKLIST

- ☑ GPO created and configured
- ☑ GPO linked to Sales OU
- ☑ gpupdate /force executed
- ☑ Control Panel access blocked
- ☑ gpresult /r shows GPO applied

KEY COMMANDS

Command	Purpose
gpupdate /force	Forces Group Policy update
gpresult /r	Shows result of applied GPOs
gpresult /h report.html	Generates HTML report
rsop.msc	Resultant Set of Policy (GUI tool)

LEARNINGS

- How to create and link a GPO
- How to configure Administrative Template settings
- How to force GPO update
- How to verify GPO application using gpresult
- How to troubleshoot inheritance issues

LAB 3: DNS ISSUE TROUBLESHOOTING

Scenario: Users are unable to access an internal application (app.company.local). You need to troubleshoot and fix the DNS issue.

1. LAB ENVIRONMENT

Domain:	company.local
DNS Server:	DC01 (192.168.1.10)
Client PC:	WIN10 (192.168.1.50)
Tool:	DNS Manager, Command Prompt

NETWORK DIAGRAM

2. ISSUE

Users cannot access the internal application using the name **app.company.local.**

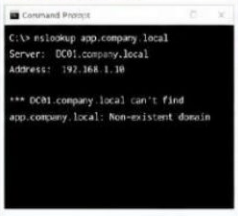
EXPECTED ERROR

3. TROUBLESHOOTING STEPS

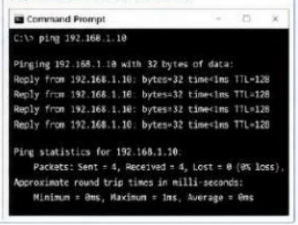
1. Open Command Prompt on client PC.
2. Test the name resolution using nslookup.
3. Verify DNS server is reachable.
4. Check DNS service on DC01.
5. Verify DNS records for app.company.local.
6. If record is missing, create A record.
7. Flush DNS cache on client.
8. Test again to confirm resolution.

STEP-BY-STEP PROCEDURE

Step 1: Test name resolution



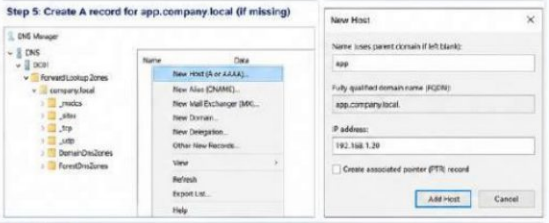
Step 2: Check DNS server reachability



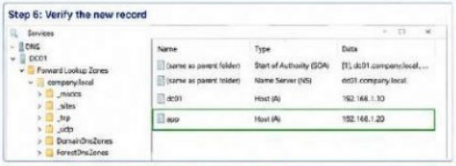
Step 3: Check DNS service on DC01

Step 4: Verify DNS records

Step 5: Create A record for app.company.local (if missing)

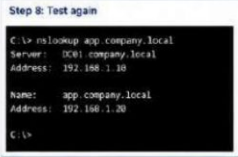


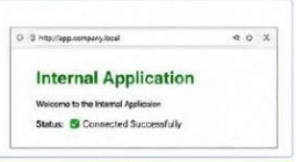
Step 6: Verify the new record



Step 7: Flush DNS cache on client

Step 8: Test again





VERIFICATION CHECKLIST

- ☑ nslookup initially failed
- ☑ DNS server reachable
- ☑ DNS service running on DC01
- ☑ A record created for app.company.local
- ☑ DNS cache flushed
- ☑ Name resolution successful

USEFUL COMMANDS

Command	Purpose
nslookup <name>	Test DNS name resolution
ping <ip>	Test connectivity to server
ipconfig /flushdns	Flush DNS cache
ipconfig /displaydns	View DNS cache
net stop dnscache	Stop DNS Client service
net start dnscache	Start DNS Client service

COMMON DNS RECORDS

Record Type	Purpose
A	Maps name to IPv4 address
AAAA	Maps name to IPv6 address
CNAME	Alias record (canonical name)
MX	Mail exchange record
NS	Name server record
PTR	Reverse lookup record

KEY POINTS

- Always verify DNS client settings.
- Check connectivity to DNS server.
- Ensure DNS service is running.
- Verify zone and records.
- Flush DNS cache after changes.
- Test again to confirm resolution.

LAB 4: DHCP FAILURE TROUBLESHOOTING

Scenario: New users are getting APIPA addresses (169.254.x.x). They cannot access network resources or internet.

1. LAB ENVIRONMENT

Domain	: company.local
DHCP Server	: DC01 (192.168.1.10)
Scope	: 192.168.1.100 – 192.168.1.200
Subnet Mask	: 255.255.255.0
Default Gateway	: 192.168.1.1
DNS Server	: 192.168.1.10
Client PC	: WIN10 (DHCP Enabled)
Tool	: DHCP Manager, Command Prompt

NETWORK DIAGRAM

DC01 / DHCP Server
192.168.1.10

WIN10 / Client
(DHCP Enabled)

2. ISSUE

New users are getting APIPA addresses (169.254.x.x) instead of valid IP addresses from DHCP.

EXPECTED ERROR

Network Connection Details

Property	Value
IPv4 Address	169.254.10.25
Subnet Mask	255.255.0.0
Default Gateway	
DHCP Server	
DNS Servers	
Lease Obtained	
Lease Expires	

3. TROUBLESHOOTING STEPS

1. Check client IP configuration.
2. Verify DHCP service is running on server.
3. Check DHCP scope for available addresses.
4. Check scope status and errors.
5. Renew IP address on client.
6. Release and renew IP address.
7. Expand DHCP scope (if needed).
8. Verify again IP configuration.
9. Test connectivity (ping gateway and DNS).

STEP 1: CHECK CLIENT IP CONFIGURATION

On Client PC (WIN10)

Command Prompt

```
C:\> ipconfig

Ethernet adapter Ethernet0:

   Connection-specific DNS Suffix . :
   Link-local IPv6 Address . . . . : fe80::9ae:ff39:b3:2f:114
   IPv4 Address. . . . . . . . . . : 169.254.10.25
   Subnet Mask . . . . . . . . . . : 255.255.0.0
   Default Gateway . . . . . . . . :

* APIPA address detected (169.254.x.x)
```

STEP 2: CHECK DHCP SERVICE ON SERVER

On DC01 (DHCP Server)

Services (Local)

Name	Status	Startup Type
DHCP Server	Running	Automatic
DNS Server	Running	Automatic
Windows Event Log	Running	Automatic
TCP/IP NetBIOS Helper	Running	Automatic

If not running, right click **DHCP Server → Start.**

STEP 3: CHECK DHCP SCOPE FOR AVAILABLE ADDRESSES

On DC01 – DHCP Manager

DHCP
- dc01.company.local
  - IPv4
    - Scope [192.168.1.0] Main Scope
      - Address Pool
      - Address Leases
      - Reservations
      - Scope Options
      - Policies
    - Server Options
    - Policies
    - Filters
  - IPv6

Address Pool

Start IP Address	End IP Address	Total Addresses	Free Addresses
192.168.1.100	192.168.1.200	101	15

Ensure there are free addresses available

STEP 4: CHECK SCOPE STATUS AND ERRORS

On DC01 – DHCP Manager

DHCP
- dc01.company.local
  - IPv4
    - Scope [192.168.1.0] Main Scope
      - Address Pool
      - Address Leases
      - Reservations
      - Scope Options
      - Policies
    - Server Options
    - Policies
    - Filters
  - IPv6

Scope [192.168.1.0] Main Scope

Status	"Active"
Start Time	5/10/2024 9:00:00 AM
End Time	Never
Subnet Mask	255.255.255.0
Free IP Addresses	15
Percentage in Use	85%
Scope State	Active

Scope is Active and working properly.

STEP 5: RENEW IP ADDRESS ON CLIENT

On Client PC (WIN10)

Command Prompt

```
C:\> ipconfig /release
Windows IP Configuration

   No operation can be performed on Ethernet0
   while it has its media disconnected.

C:\> ipconfig /renew

Windows IP Configuration

   An error occurred while renewing interface
   Ethernet0 : Unable to contact your DHCP server.
```

STEP 6: RELEASE AND RENEW IP ADDRESS

On Client PC (WIN10)

Command Prompt

```
C:\> ipconfig /release
C:\> ipconfig /renew
Windows IP Configuration

Ethernet adapter Ethernet0:
   IPv4 Address. . . . . . . . . . : 192.168.1.120
   Subnet Mask . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . : 192.168.1.10
   DNS Servers . . . . . . . . . . : 192.168.1.10

* Successfully obtained IP from DHCP
```

STEP 7: EXPAND DHCP SCOPE (IF NEEDED)

On DC01 – DHCP Manager

Address Pool

Start IP Address	End IP Address
192.168.1.100	192.168.1.200

To expand:

1. Right click **Address Pool**
2. Select **Configure Options**
3. Increase **End IP Address** (e.g., 192.168.1.250)
4. Click **OK**

STEP 8: VERIFY IP CONFIGURATION AGAIN

On Client PC (WIN10)

Command Prompt

```
C:\> ipconfig
Ethernet adapter Ethernet0:
   IPv4 Address. . . . . . . . . . : 192.168.1.120
   Subnet Mask . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . : 192.168.1.1
   DHCP Server . . . . . . . . . . : 192.168.1.10
   DNS Servers . . . . . . . . . . : 192.168.1.10
   Lease Obtained . . . . . . . . . : 5/10/2024 10:15:22 AM
   Lease Expires . . . . . . . . . . : 5/10/2024 2:15:22 PM
```

STEP 9: TEST CONNECTIVITY

On Client PC (WIN10)

```
C:\> ping 192.168.1.1
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64
Reply from 192.168.1.1: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\> ping 192.168.1.10
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

* Connectivity successfull
```

COMMON CAUSES AND FIXES

Issue	Fix
DHCP service stopped	Start DHCP Server service
No IP addresses in scope	Add or expand address pool
Scope inactive	Activate the scope
DHCP server not authorized	Authorize DHCP server in AD
Network connectivity issue	Check cables, switches, VLAN
Client firewall blocking DHCP	Allow DHCP (UDP 67,68)
IP conflict	Check duplicates in address leases

VERIFICATION CHECKLIST

- ☑ Client gets IP from DHCP (not APIPA)
- ☑ DHCP service is running
- ☑ Scope has available addresses
- ☑ Client can ping gateway and DNS
- ☑ Internet access is working

USEFUL COMMANDS

Command	Purpose
ipconfig /all	View full IP configuration
ipconfig /release	Release current IP
ipconfig /renew	Renew IP from DHCP
ipconfig /flushdns	Flush DNS cache
ping <ip>	Test connectivity

DHCP PORTS (FOR REFERENCE)

Port	Protocol	Purpose
67	UDP	DHCP Server
68	UDP	DHCP Client
547	UDP	DHCPv6 Server
546	UDP	DHCPv6 Client

KEY POINTS

- APIPA (169.254.x.x) means client could not reach DHCP server.
- Always check DHCP service and scope before troubleshooting.
- Release and renew IP after making changes.
- Ensure clients can reach DHCP server on the network.
- Monitor DHCP logs in Event Viewer for errors.

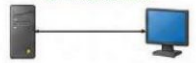
LAB 5: FILE SERVER PERMISSION ISSUE

Scenario: Finance department users cannot access the shared folder \\FS01\Finance. Fix the share and NTFS permissions so only Finance users can access it.

1. LAB ENVIRONMENT

Domain	: company.local
File Server	: FS01 (192.168.1.10)
Client PC	: WIN10 (192.168.1.50)
Share Name	: Finance
Share Path	: D:\Shares\Finance
Domain Group	: Finance_Users
Test User	: john.finance (member of Finance_Users)
Tool	: File Explorer, Computer Management

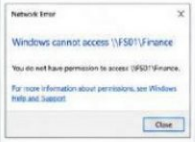
NETWORK DIAGRAM



2. ISSUE

Users in Finance_Users group receive "Access is denied" when trying to access \\FS01\Finance.

EXPECTED ERROR

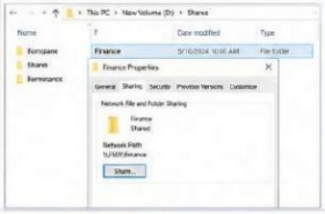


3. STEPS OVERVIEW

1. Create shared folder on the server.
2. Configure Share Permissions.
3. Configure NTFS Permissions.
4. Remove unnecessary permissions.
5. Test access from client.
6. Verify only Finance users can access.

4. CREATE SHARED FOLDER ON SERVER (FS01)

1. On FS01, open File Explorer.
2. Go to D:\Shares.
3. Right click → New → Folder.
4. Name it: **Finance**.



5. CONFIGURE SHARE PERMISSIONS

1. Right click Finance folder → Properties.
2. Go to Sharing tab → Click **Share**.
3. Add group: **Finance_Users**
4. Permission Level: **Change** → **Read/Write**
5. Click **Share** → **Done**.

6. CONFIGURE NTFS PERMISSIONS

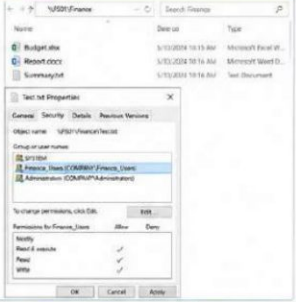
1. Go to Security tab → Edit.
2. Remove Users, Everyone and other groups.
3. Click Add → Advanced.
4. Find Now → Select Finance_Users → OK.
5. Select Finance_Users and allow: Modify
6. Click Apply → OK.

7. REMOVE UNNECESSARY PERMISSIONS

1. Select Users → Remove.
2. Select Everyone → Remove.
3. Keep only:
   - SYSTEM
   - Administrators
   - Finance_Users
4. Ensure **Finance_Users** has Modify permission.

8. TEST ACCESS FROM CLIENT (WIN10)

1. On WIN10, press Windows + R, type \\FS01\Finance and press Enter.
2. Login with john.finance (member of Finance_Users).
3. You should be able to open the folder and create files.



9. VERIFY RESTRICTIONS

1. Login with any user NOT in Finance_Users.
2. Try to access \\FS01\Finance
3. You should see:

VERIFICATION CHECKLIST

- ☑ Share \\FS01\Finance is created
- ☑ Share Permissions set for Finance_Users
- ☑ NTFS Permissions set for Finance_Users (Modify)
- ☑ Unnecessary permissions removed
- ☑ Users in Finance_Users can access
- ☑ Users not in group cannot access

USEFUL COMMANDS

Command	Purpose
\\FS01\Finance	Open the shared folder from client
icacls "D:\Shares\Finance"	View NTFS permissions
icacls "D:\Shares\Finance" /grant COMPANY\Finance_Users:(OI)(CI)M	Grant Modify permission to group
net share	List all shares on the server
net share Finance /delete	Delete the share

SHARE PERMISSION LEVELS

Permission	Description
Full Control	Read, Write, Change & Delete
Change	Read, Write, Delete (no change permissions)
Read	View files and folders
Read/Write	View and write files
Special	Custom permissions

KEY POINTS

- Use Share Permissions to control access over the network.
- Use NTFS Permissions to secure the actual data.
- Always follow least privilege principle.
- Remove 'Users' and 'Everyone' for security.
- Test with both allowed and denied users.
- Document permission changes.

LAB 6: PRINT SERVER TROUBLESHOOTING

Issue: Users cannot print. The printer shows as "Offline". Fix the printer, clear the queue, reinstall the driver if needed and deploy via GPO.

1. LAB ENVIRONMENT

Domain	: company.local
Print Server	: PS01 (192.168.1.20)
Printer Name	: HP LaserJet P2055
Share Name	: HP2055
Client PC	: WIN10 (192.168.1.50)
User	: john.finance (member of Finance_Users)
Tool	: Print Management, Services, GPO

NETWORK DIAGRAM

2. ISSUE

Users are unable to print.
The printer shows as "Offline" on the client.

Devices and Printers (on client)

HP LaserJet P2055
Offline

EXPECTED BEHAVIOR

Printer should show as "Ready" and documents should print successfully.

3. OVERVIEW OF STEPS

1. Check printer and server connectivity.
2. Check Print Spooler service.
3. Clear the print queue.
4. Reinstall or update the printer driver (if needed).
5. Test printing from client.
6. Deploy printer to users via GPO.
7. Verify from another client.

4. CHECK PRINTER CONNECTIVITY

On client PC, open Command Prompt and run:

```
Command Prompt
C:\> ping 192.168.1.20

Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>
```

If ping fails, fix network issues first.

5. CHECK PRINT SPOOLER SERVICE ON PS01

On PS01 server:

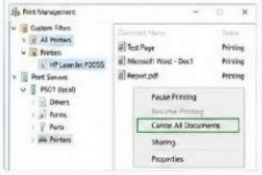
1. Press Win + R → type services.msc → Enter.
2. Find "Print Spooler".
3. Ensure Status is "Running" and Startup type is "Automatic".

Name	Status	Startup Type
Print Spooler	Running	Automatic
Server	Running	Automatic
Workstation	Running	Automatic

If not running:
Right click → Start.
If stopped frequently, check Event Viewer.

6. CLEAR PRINT QUEUE

1. On PS01, open Print Management.
2. Go to: Print Management → Custom Filters → All Printers.
3. Right click on HP LaserJet P2055 → See what's printing.
4. Right click → Cancel All Documents.



7. REINSTALL / UPDATE PRINTER DRIVER (IF NEEDED)

1. On PS01, open Print Management.
2. Expand Print Servers → PS01 (local) → Drivers.
3. Right click HP LaserJet P2055 driver → Remove Driver Package.
4. Download latest driver from HP website.
5. Install driver:
   Right click Drivers → Add Driver →
   Choose the downloaded .inf file → Next.
6. Re-add the printer (if removed).

8. TEST PRINTING FROM CLIENT

1. On WIN10 client, open Devices and Printers.
2. Right click HP LaserJet P2055 → See what's printing (should be empty).
3. Click Printer → Print Test Page.
4. Printer should be "Ready" and print the page.

9. DEPLOY PRINTER VIA GPO

1. Open Group Policy Management (gpmc.msc).
2. Create or edit a GPO and link to Finance_Users OU.
3. Go to:
   User Configuration → Preferences → Control Panel Settings → Printers.
4. Right click Printers → New → Shared Printer.
5. Path: \\PS01\HP2055
6. Action: Update. Check "Reapply once then remove item" (optional).
7. Click OK and Close.

10. VERIFY FROM ANOTHER CLIENT

1. Log on to another client or run: gpupdate /force
2. Check Devices and Printers.
3. HP LaserJet P2055 should appear automatically.
4. Print a test page and confirm.

11. USEFUL COMMANDS

Command	Purpose
services.msc	Open Services
net stop spooler	Stop Print Spooler service
net start spooler	Start Print Spooler service
net stop spooler & del %systemroot% System32\spool\PRINTERS*.*	Clear print spool (stop service first)
net start spooler	Start Print Spooler service
gpupdate /force	Force Group Policy update
rundll32 printui.dll,PrintUIEntry /y /n "HP LaserJet P2055" /q	Add local printer via command line

12. COMMON CAUSES AND FIXES

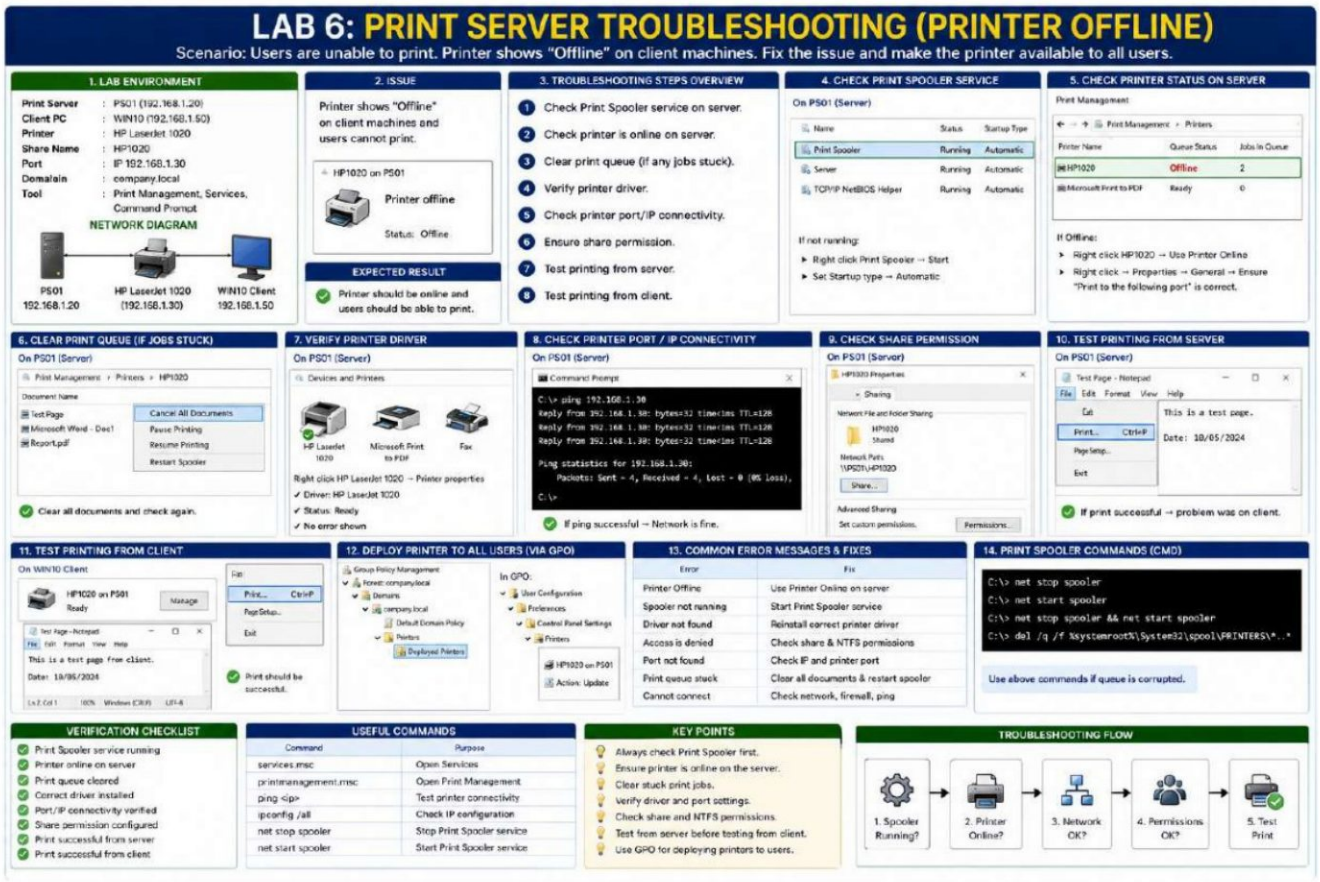
Cause	Fix
Print Spooler is stopped	Start and set to Automatic
Documents stuck in queue	Clear print queue
Invalid or corrupt driver	Reinstall / update driver
Network issue	Check connectivity / firewall
Permission issue	Check share and NTFS permissions
Wrong port	Use correct port (Standard TCP/IP port)
GPO not applied	Run gpupdate /force and verify link

VERIFICATION CHECKLIST

- ☑ Printer is reachable (ping successful)
- ☑ Print Spooler service is running
- ☑ Print queue is clear
- ☑ Correct driver is installed
- ☑ Client can print test page
- ☑ Printer deployed via GPO
- ☑ Another client can print successfully

KEY POINTS

- Always check Spooler service first.
- Clearing queue resolves most issues.
- Use latest and correct driver.
- Test from multiple clients.
- GPO deployment saves time and ensures consistency.
- Check Event Viewer for detailed errors if issue persists.


LAB 6: PRINT SERVER TROUBLESHOOTING (PRINTER OFFLINE)
Scenario: Users are unable to print. Printer shows "Offline" on client machines. Fix the issue and make the printer available to all users.
1. LAB ENVIRONMENT
Print Server : PS01 (192.168.1.20)
Client PC : WIN10 (192.168.1.50)
Printer : HP Laserjet 1020
Share Name : HP1020
Port : IP 192.168.1.30
Domain : company.local
Tool : Print Management, Services, Command Prompt
NETWORK DIAGRAM
PS01 192.168.1.20
HP Laserjet 1020 (192.168.1.30)
WIN10 Client 192.168.1.50
2. ISSUE
Printer shows "Offline" on client machines and users cannot print.
HP1020 on PS01
Printer offline
Status: Offline
EXPECTED RESULT
Printer should be online and users should be able to print.
3. TROUBLESHOOTING STEPS OVERVIEW
1 Check Print Spooler service on server.
2 Check printer is online on server.
3 Clear print queue (if any jobs stuck).
4 Verify printer driver.
5 Check printer port/IP connectivity.
6 Ensure share permission.
7 Test printing from server.
8 Test printing from client.
4. CHECK PRINT SPOOLER SERVICE
On PS01 (Server)
Name Status Startup Type
Print Spooler Running Automatic
Server Running Automatic
TCP/IP NetBIOS Helper Running Automatic
If not running:
Right click Print Spooler → Start
Set Startup type → Automatic
5. CHECK PRINTER STATUS ON SERVER
Print Management
Printer Name Queue Status Jobs in Queue
HP1020 Offline 2
Microsoft Print to PDF Ready 0
If Offline:
Right click HP1020 → Use Printer Online
Right click → Properties → General → Ensure "Print to the following port" is correct.
6. CLEAR PRINT QUEUE (IF JOBS STUCK)
On PS01 (Server)
Document Name
Test Page
Microsoft Word - Doc1
Report.pdf
Cancel All Documents
Pause Printing
Resume Printing
Restart Spooler
Clear all documents and check again.
7. VERIFY PRINTER DRIVER
On PS01 (Server)
Devices and Printers
HP Laserjet 1020
Microsoft Print to PDF
Fax
Right click HP LaserJet 1020 → Printer properties
✓ Driver: HP LaserJet 1020
✓ Status: Ready
✓ No error shown
8. CHECK PRINTER PORT / IP CONNECTIVITY
On PS01 (Server)
Command Prompt
C:\> ping 192.168.1.30
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Reply from 192.168.1.30: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.30:
Packets: Sent = 4, Received = 4, Lost = 0 (0% Loss),
C:\>
If ping successful → Network is fine.
9. CHECK SHARE PERMISSION
On PS01 (Server)
HP1020 Properties
Sharing
Network File and Folder Sharing
HP1020
Shared
Network Path:
\\PS01\HP1020
Share...
Advanced Sharing
Set custom permissions.
Permissions...
10. TEST PRINTING FROM SERVER
On PS01 (Server)
Test Page - Notepad
File Edit Format View Help
Print... Ctrl+P
This is a test page.
Date: 10/05/2024
If print successful → problem was on client.
11. TEST PRINTING FROM CLIENT
On WIN10 Client
HP1020 on PS01
Ready
Manage
Print... Ctrl+P
This is a test page from client.
Date: 10/05/2024
Print should be successful.
12. DEPLOY PRINTER TO ALL USERS (VIA GPO)
Group Policy Management
Forest: company.local
Domains
company.local
Default Domain Policy
Printers
Deployed Printers
In GPO:
User Configuration
Preferences
Control Panel Settings
Printers
HP1020 on PS01
Action: Update
13. COMMON ERROR MESSAGES & FIXES
Error Fix
Printer Offline Use Printer Online on server
Spooler not running Start Print Spooler service
Driver not found Reinstall correct printer driver
Access is denied Check share & NTFS permissions
Port not found Check IP and printer port
Print queue stuck Clear all documents & restart spooler
Cannot connect Check network, firewall, ping
14. PRINT SPOOLER COMMANDS (CMD)
C:\> net stop spooler
C:\> net start spooler
C:\> net stop spooler && net start spooler
C:\> del /q /f %systemroot%\System32\spool\PRINTERS*.*
Use above commands if queue is corrupted.
VERIFICATION CHECKLIST
Print Spooler service running
Printer online on server
Print queue cleared
Correct driver installed
Port/IP connectivity verified
Share permission configured
Print successful from server
Print successful from client
USEFUL COMMANDS
Command Purpose
services.msc Open Services
printmanagement.msc Open Print Management
ping <ip> Test printer connectivity
ipconfig /all Check IP configuration
net stop spooler Stop Print Spooler service
net start spooler Start Print Spooler service
KEY POINTS
Always check Print Spooler first.
Ensure printer is online on the server.
Clear stuck print jobs.
Verify driver and port settings.
Check share and NTFS permissions.
Test from server before testing from client.
Use GPO for deploying printers to users.
TROUBLESHOOTING FLOW
1. Spooler Running?
2. Printer Online?
3. Network OK?
4. Permissions OK?
5. Test Print

LAB 7: OUTLOOK ISSUE – OUTLOOK KEEPS ASKING FOR PASSWORD

Scenario: Outlook keeps asking for password repeatedly. Fix the issue by recreating profile, clearing credentials, checking Autodiscover and repairing OST.

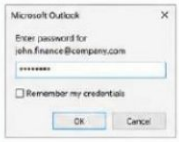
1. LAB ENVIRONMENT

Domain	: company.local
Mail System	: Microsoft 365 (O365)
Client PC	: WIN10 (192.168.1.50)
User	: john.finance@company.com
Tool	: Outlook, Control Panel, CMD, Credential Manager, Outlook ScanPST

NETWORK DIAGRAM

2. ISSUE

User opens Outlook.
Outlook keeps asking for password.



EXPECTED BEHAVIOR

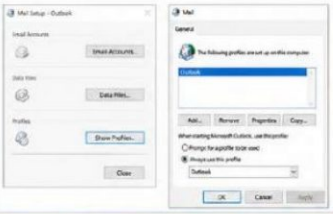
Outlook should connect without asking password.

3. OVERVIEW OF STEPS

1. Recreate Outlook profile.
2. Clear saved credentials.
3. Check Autodiscover.
4. Repair OST file.
5. Test Outlook connectivity.
6. Verify and confirm fix.

4. [illegible]

1. Press Windows + R, type control mlcfg32.cpl and press Enter.
2. Click Show Profiles.
3. Click Add, give profile name (e.g., Outlook).
4. Add email account and complete setup.
5. Set the new profile as "Always use this profile".
6. Click Apply and OK.

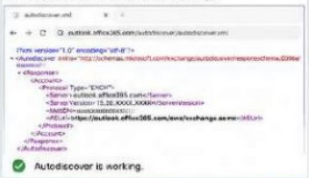


5. CLEAR SAVED CREDENTIALS

1. Open Control Panel → Credential Manager.
2. Click Windows Credentials.
3. Find entries related to:
   - Outlook
   - MicrosoftOffice16_Data
   - company.com (or your domain)
4. Click Remove for each related entry.
5. Restart Outlook.

6. CHECK AUTODISCOVER

1. Open browser and go to:
   https://outlook.office365.com/autodiscover/autodiscover.xml
2. Sign in with your email and password.
3. You should see XML response.
4. If error, check DNS, Internet or proxy settings.



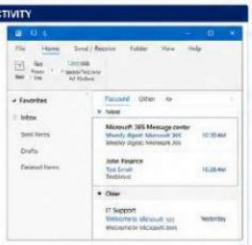
Autodiscover is working.

7. REPAIR OST FILE

1. Close Outlook.
2. Open ScanPST.exe (usually in: C:\Program Files\Microsoft Office\root\Office16).
3. Browse and select the OST file.
4. Click Start to scan.
5. Click Repair if issues are found.
6. Wait until repair is completed.
7. Restart Outlook.

8. TEST OUTLOOK CONNECTIVITY

1. Open Outlook.
2. Enter password once when prompted.
3. Outlook should open without asking password again.
4. Send/Receive test email.



Outlook is working correctly.

9. VERIFY AND CONFIRM FIX

- Outlook opens without asking password.
- Can send and receive emails.
- Calendar, Contacts and Tasks sync properly.
- No repeated password prompts.
- Issue resolved.

VERIFICATION CHECKLIST

- Outlook profile recreated successfully
- Old credentials removed
- Autodiscover working
- OST file repaired (if needed)
- Outlook opens without password prompt
- Email send/receive working
- Issue resolved and confirmed

USEFUL COMMANDS / PATHS

Command / Path	Purpose
control mlcfg32.cpl	Open Mail (Profiles)
rundll32.exe keymgr.dll, KRShowKeyMgr	Open Credential Manager
ping outlook.office365.com	Check connectivity
ipconfig /flushdns	Flush DNS cache
C:\Program Files\Microsoft Office\root\Office16\SCANPST.EXE	Outlook Inbox Repair Tool

COMMON CAUSES AND FIXES

Cause	Fix
Corrupt Outlook profile	Recreate Outlook profile
Saved wrong credentials	Clear Credential Manager
Autodiscover not working	Check URL/DNS/Proxy
Corrupt OST file	Run ScanPST and repair
Network / MFA issues	Check Internet and MFA

KEY POINTS

- Always start with profile recreation.
- Remove old/incorrect credentials.
- Verify Autodiscover in browser.
- Repair OST if Outlook data file is corrupted.
- Test email functionality after each step.
- Document the actions and resolution.

LAB 8: OFFICE 365 USER ADMINISTRATION

Scenario: Perform day-to-day user administration tasks in Microsoft 365 admin center.

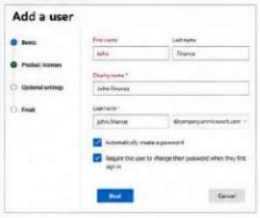
1. LAB ENVIRONMENT

Tenant Name	: company.onmicrosoft.com
Admin Account	: admin@company.onmicrosoft.com
Platform	: Microsoft 365 Admin Center
Client PC	: WIN10 (192.168.1.50)
Tool	: Microsoft 365 admin center
URL	: https://admin.microsoft.com

NETWORK DIAGRAM

2. CREATE USER WITH MAILBOX

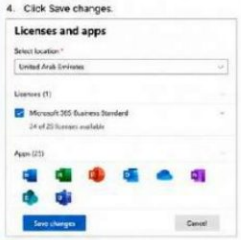
1. Go to Microsoft 365 admin center.
2. Users > Active users > Add a user.
3. Add name, username, domain and password.
4. Assign product license.
5. Click Add user.



User john.finance@company.onmicrosoft.com created.

3. ASSIGN LICENSE

1. In Active users, select the user.
2. Click Licenses and apps.
3. Select a license (e.g., Microsoft 365 Business Standard).
4. Click Save changes.



License assigned successfully.

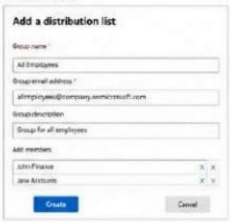
4. CREATE SHARED MAILBOX

1. Go to Teams & groups > Shared mailboxes.
2. Click Add a shared mailbox.
3. Enter name and email address.
4. Add owners and members.
5. Click Create.

Shared mailbox finance@company.onmicrosoft.com created.

5. CREATE DISTRIBUTION GROUP

1. Go to Teams & groups > Groups.
2. Click Add a group > Distribution list.
3. Enter name and description.
4. Add members.
5. Click Create.



Distribution group created successfully.

6. PASSWORD RESET

1. Go to Active users.
2. Select the user.
3. Click Reset password.
4. Choose to Auto-generate password or enter a new password.
5. Check "Require this user to change password when they next sign in".
6. Click Reset password.

Password reset successfully.

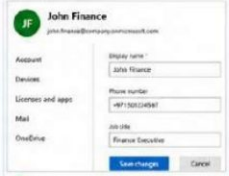
7. ADD USER TO GROUP

1. Go to Teams & groups > Groups.
2. Select the group (e.g., All Employees).
3. Click Edit.
4. Add members.
5. Click Save

User added to the group.

8. MODIFY USER PROPERTIES

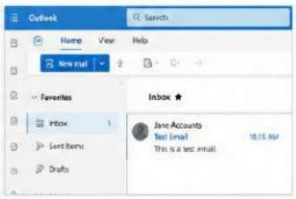
1. Go to Active users.
2. Select the user.
3. Click Manage username, profile or roles.
4. Update properties (e.g., display name, phone).
5. Click Save changes.



User properties updated.

9. VERIFY MAILBOX AND SIGN-IN

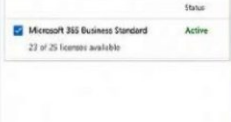
1. Sign in as the user at https://outlook.office.com
2. Verify mailbox access.
3. Send and receive a test email.



Mailbox access verified successfully.

10. VERIFY LICENSE ASSIGNMENT

1. Go to Active users.
2. Select the user.
3. Under Licenses and apps, verify the license status is "Active".



License is active for the user.

VERIFICATION CHECKLIST

- ☑ User with mailbox created
- ☑ License assigned and active
- ☑ Shared mailbox created
- ☑ Distribution group created
- ☑ Password reset successfully
- ☑ User added to group
- ☑ Mailbox login and email test successful

USEFUL ADMIN CENTER PATHS

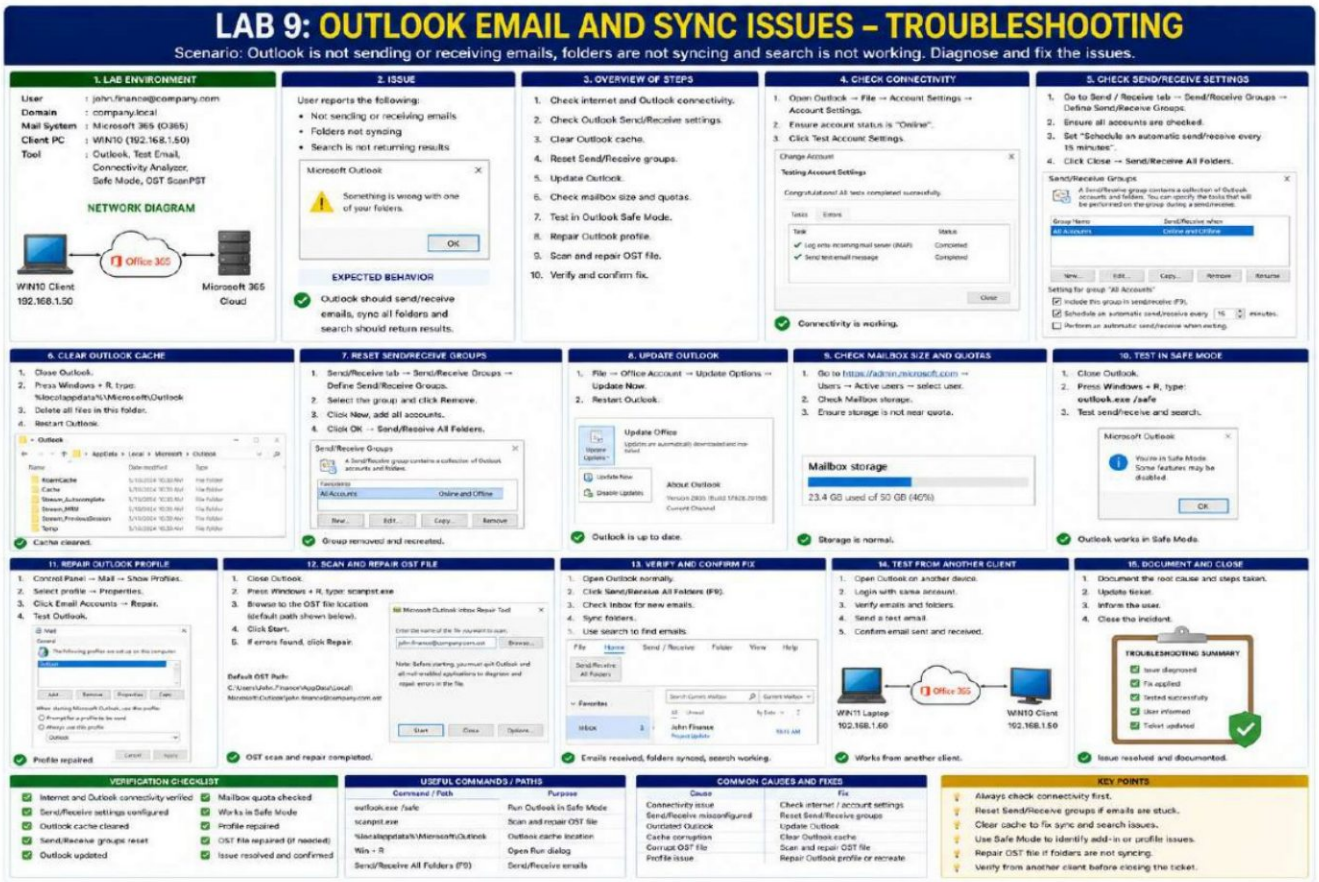
Task	Path
Active users	Users > Active users
Add user	Users > Active users > Add a user
Licenses	Users > Active users > Licenses and apps
Shared mailbox	Teams & groups > Shared mailboxes
Groups	Teams & groups > Groups
Reset password	Users > Active users > Reset password

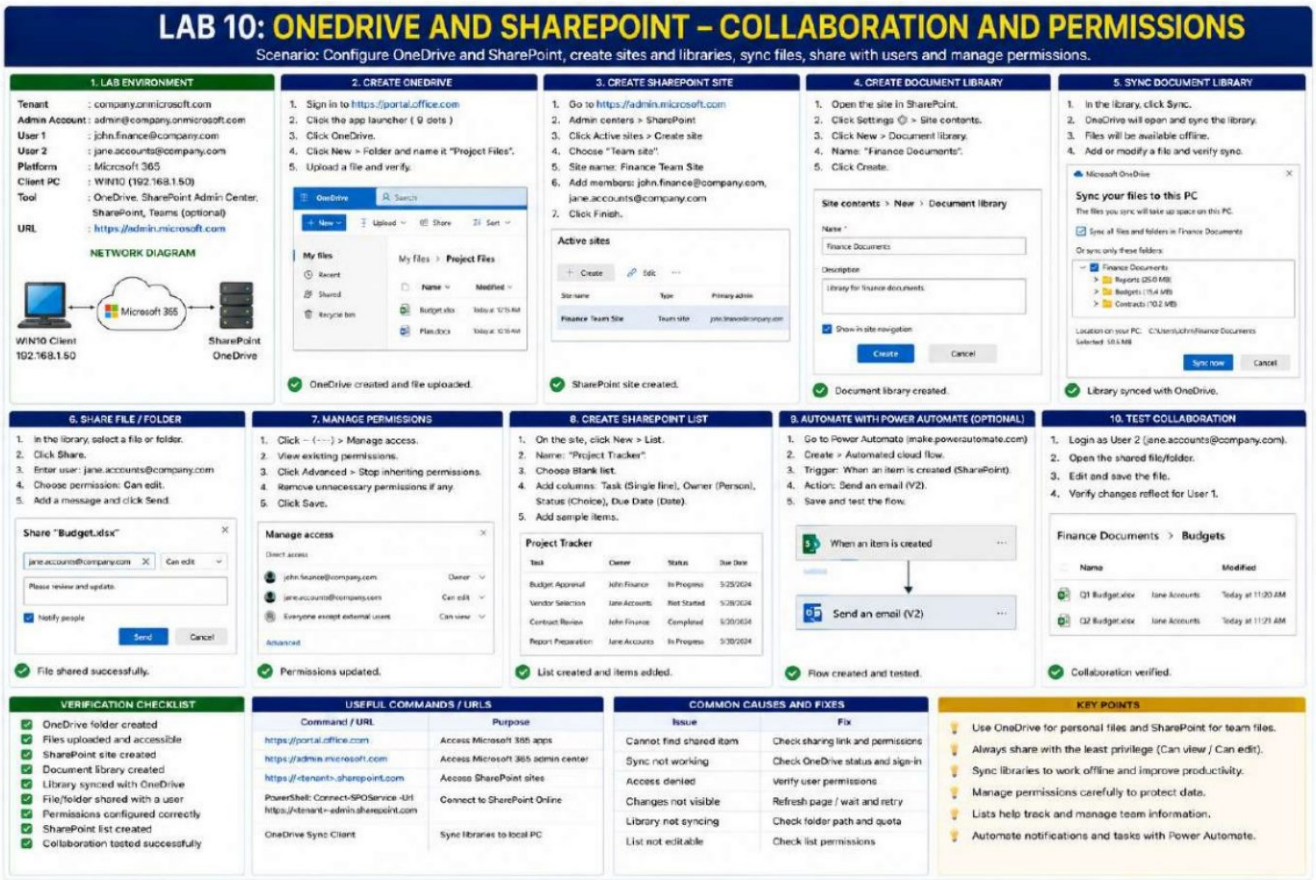
COMMON CAUSES AND FIXES

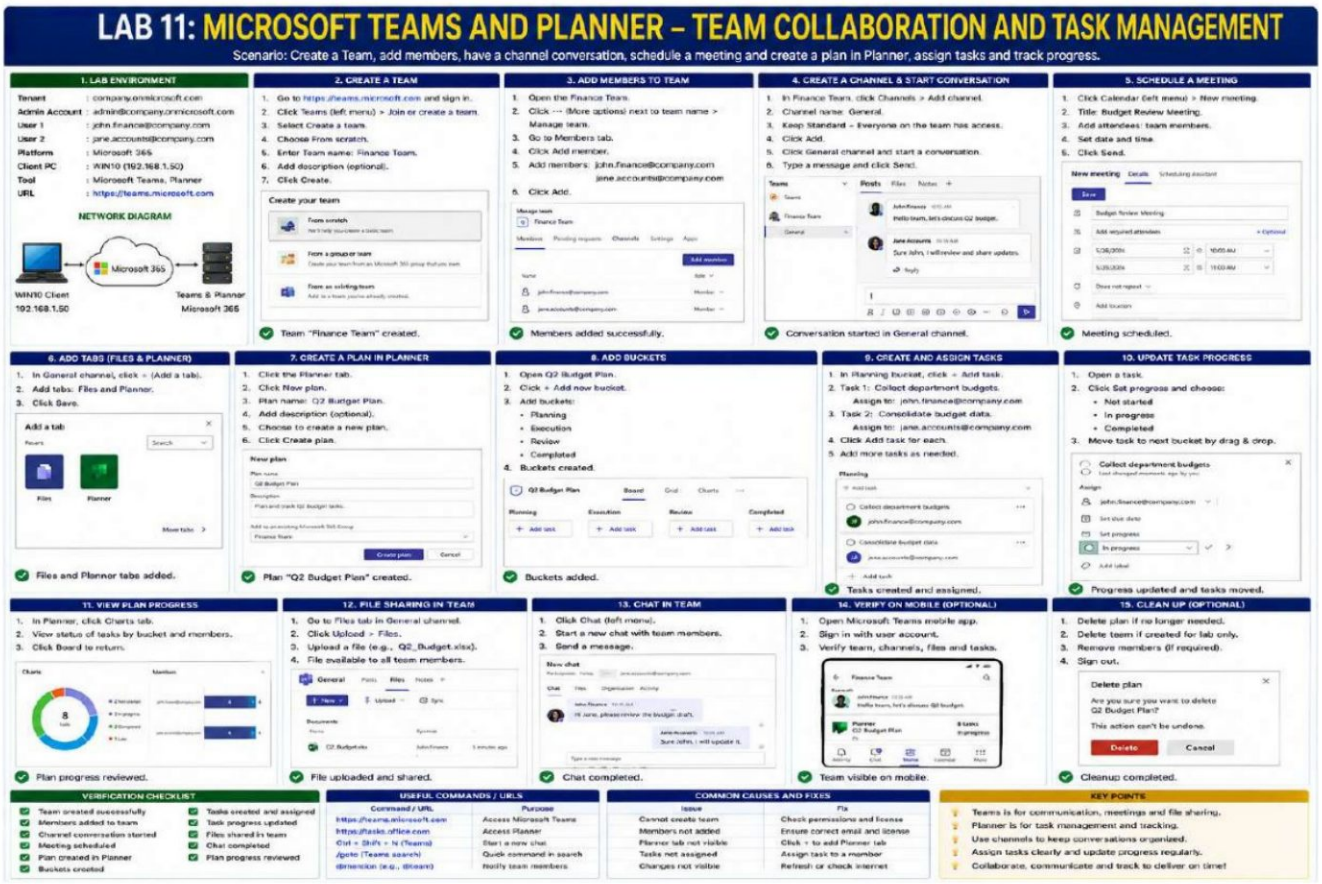
Issue	Fix
License not assigned	Assign a valid license
Cannot sign in	Check password and license
Mailbox not found	Wait a few minutes and retry
Group not showing	Refresh or check membership
Email not received	Check spam/junk and send again

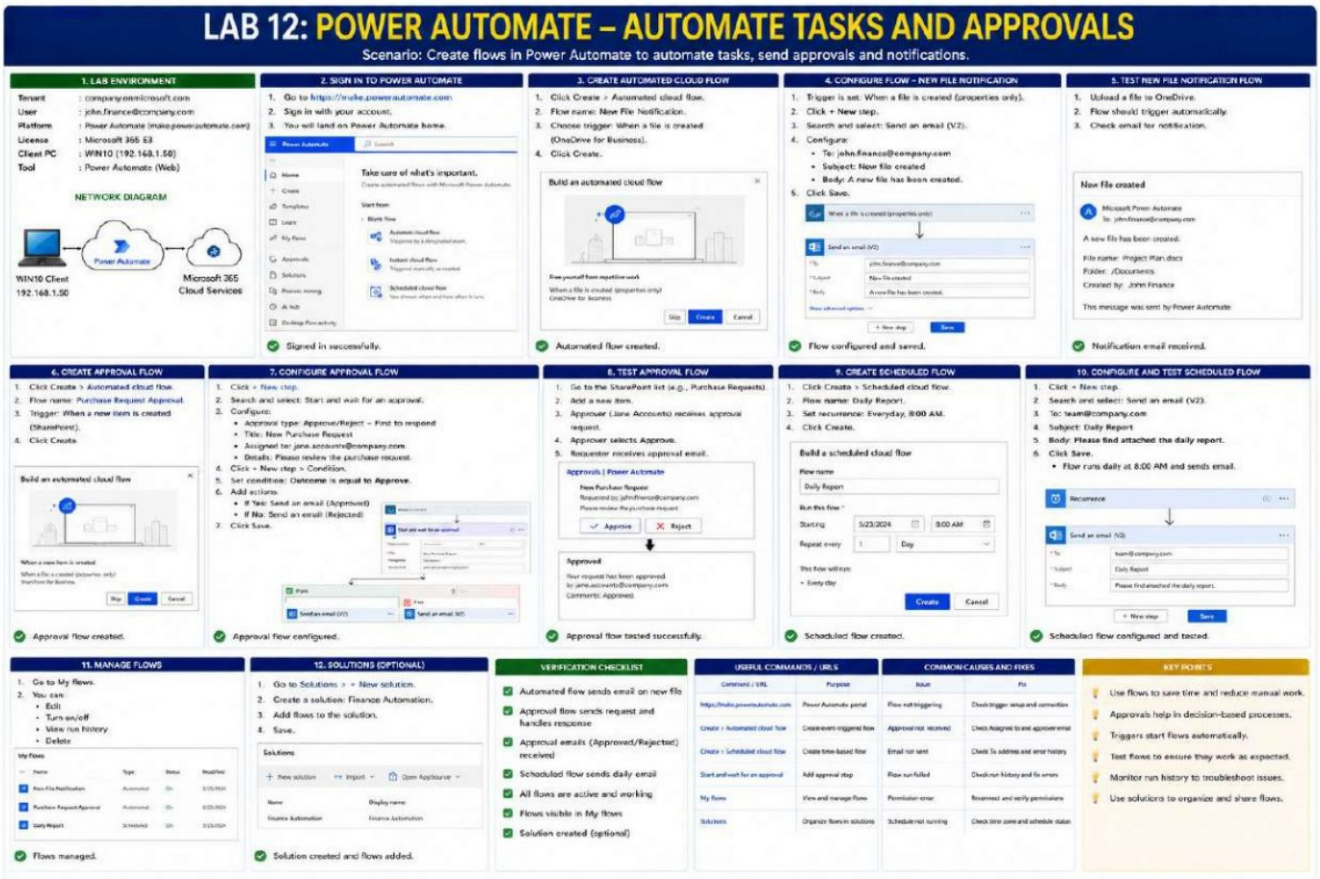
KEY POINTS

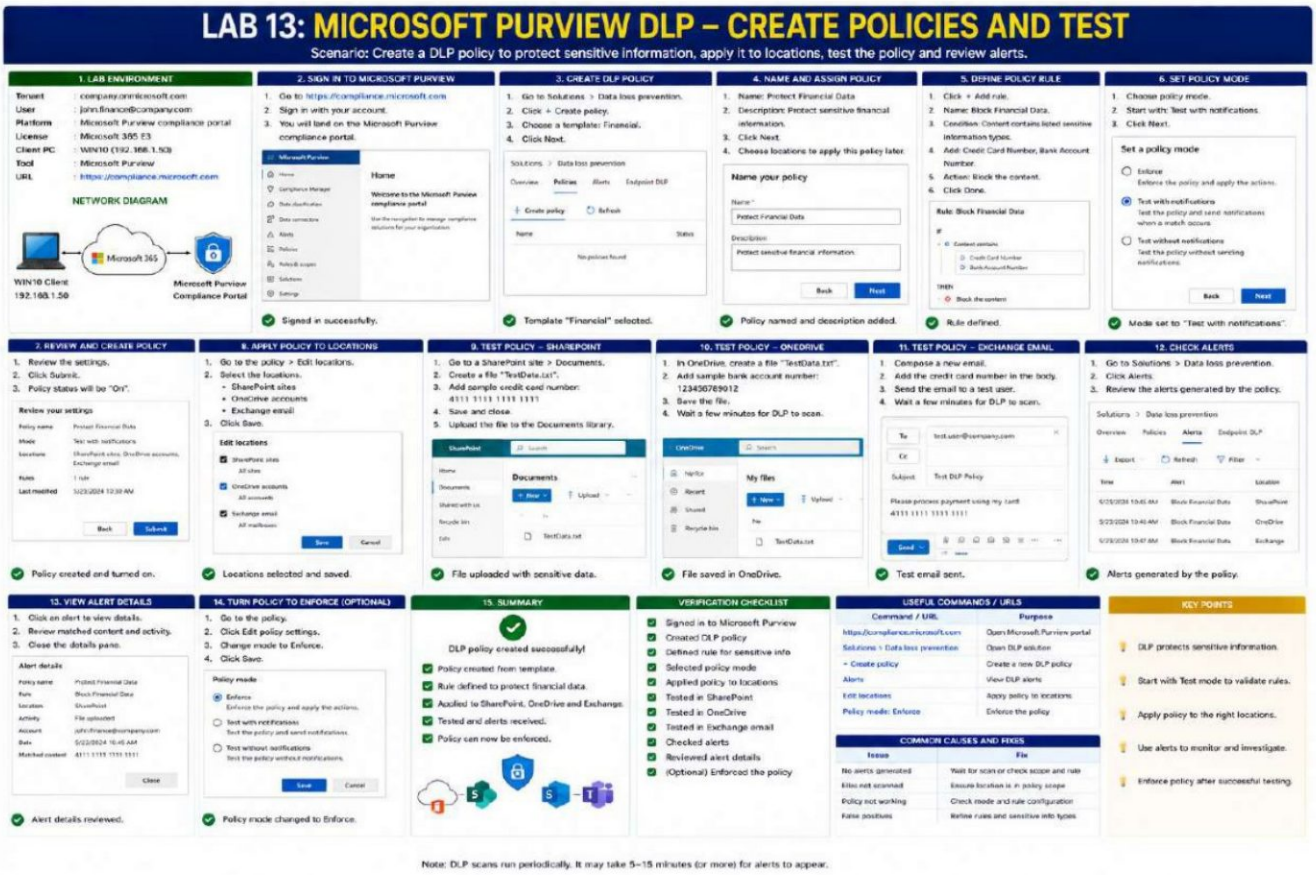
- Always assign a license to enable services.
- Use strong passwords and enforce change at first sign-in.
- Use shared mailboxes for team access.
- Use distribution groups to send emails to multiple users.
- Changes may take a few minutes to reflect.

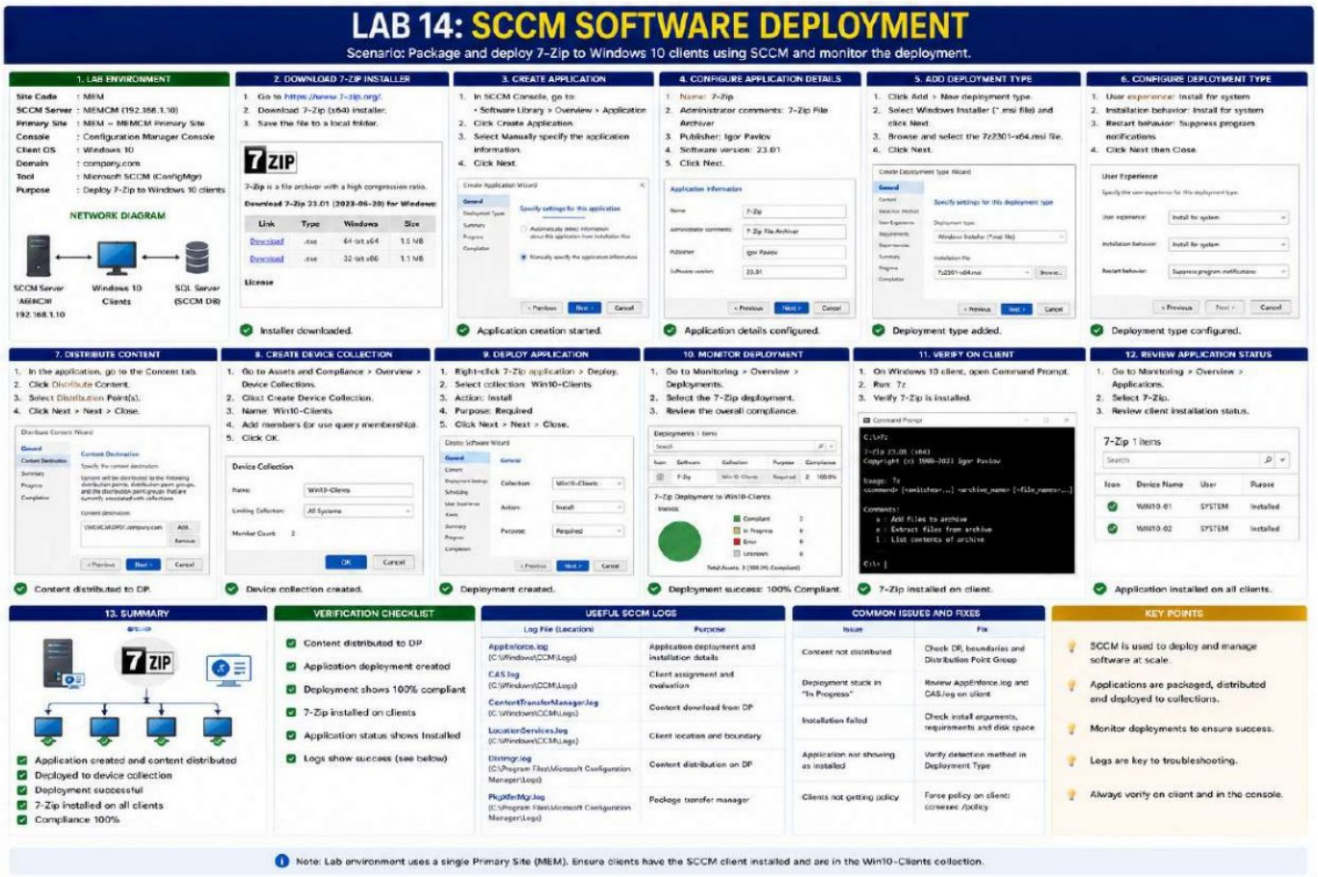

LAB 9: OUTLOOK EMAIL AND SYNC ISSUES – TROUBLESHOOTING
Scenario: Outlook is not sending or receiving emails, folders are not syncing and search is not working. Diagnose and fix the issues.
1. LAB ENVIRONMENT
User : john.finance@company.com
Domain : company.local
Mail System : Microsoft 365 (O365)
Client PC : WIN10 (192.168.1.50)
Tool : Outlook, Test Email, Connectivity Analyzer, Safe Mode, OST ScanPST
NETWORK DIAGRAM
Office 365
WIN10 Client 192.168.1.50
Microsoft 365 Cloud
2. ISSUE
User reports the following:
• Not sending or receiving emails
• Folders not syncing
• Search is not returning results
Microsoft Outlook
Something is wrong with one of your folders.
OK
EXPECTED BEHAVIOR
Outlook should send/receive emails, sync all folders and search should return results.
3. OVERVIEW OF STEPS
1. Check internet and Outlook connectivity.
2. Check Outlook Send/Receive settings.
3. Clear Outlook cache.
4. Reset Send/Receive groups.
5. Update Outlook.
6. Check mailbox size and quotas.
7. Test in Outlook Safe Mode.
8. Repair Outlook profile.
9. Scan and repair OST file.
10. Verify and confirm fix.
4. CHECK CONNECTIVITY
1. Open Outlook → File → Account Settings → Account Settings.
2. Ensure account status is "Online".
3. Click Test Account Settings.
Connectivity is working.
5. CHECK SEND/RECEIVE SETTINGS
1. Go to Send / Receive tab → Send/Receive Groups → Define Send/Receive Groups.
2. Ensure all accounts are checked.
3. Set "Schedule an automatic send/receive every 15 minutes".
4. Click Close → Send/Receive All Folders.
6. CLEAR OUTLOOK CACHE
1. Close Outlook.
2. Press Windows + R, type: %localappdata%\Microsoft\Outlook
3. Delete all files in this folder.
4. Restart Outlook.
Cache cleared.
7. RESET SEND/RECEIVE GROUPS
1. Send/Receive tab → Send/Receive Groups → Define Send/Receive Groups.
2. Select the group and click Remove.
3. Click New, add all accounts.
4. Click OK → Send/Receive All Folders.
Group removed and recreated.
8. UPDATE OUTLOOK
1. File → Office Account → Update Options → Update Now.
2. Restart Outlook.
Outlook is up to date.
9. CHECK MAILBOX SIZE AND QUOTAS
1. Go to https://admin.microsoft.com → Users → Active users → select user.
2. Check Mailbox storage.
3. Ensure storage is not near quota.
Mailbox storage
23.4 GB used of 50 GB (46%)
Storage is normal.
10. TEST IN SAFE MODE
1. Close Outlook.
2. Press Windows + R, type: outlook.exe /safe
3. Test send/receive and search.
Microsoft Outlook
You're in Safe Mode. Some features may be disabled.
OK
Outlook works in Safe Mode.
11. REPAIR OUTLOOK PROFILE
1. Control Panel → Mail → Show Profiles.
2. Select profile → Properties.
3. Click Email Accounts → Repair.
4. Test Outlook.
Profile repaired
12. SCAN AND REPAIR OST FILE
1. Close Outlook.
2. Press Windows + R, type: scanpst.exe
3. Browse to the OST file location (default path shown below).
4. Click Start.
5. If errors found, click Repair.
Default OST Path:
C:\Users\John.Finance\AppData\Local\Microsoft\Outlook\john.finance@company.com.ost
OST scan and repair completed.
13. VERIFY AND CONFIRM FIX
1. Open Outlook normally.
2. Click Send/Receive All Folders (F9).
3. Check Inbox for new emails.
4. Sync folders.
5. Use search to find emails.
Emails received, folders synced, search working.
14. TEST FROM ANOTHER CLIENT
1. Open Outlook on another device.
2. Login with same account.
3. Verify emails and folders.
4. Send a test email.
5. Confirm email sent and received.
Office 365
WIN11 Laptop 192.168.1.60
WIN10 Client 192.168.1.50
Works from another client.
15. DOCUMENT AND CLOSE
1. Document the root cause and steps taken.
2. Update ticket.
3. Inform the user.
4. Close the incident.
TROUBLESHOOTING SUMMARY
Issue diagnosed
Fix applied
Tested successfully
User informed
Ticket updated
Issue resolved and documented.
VERIFICATION CHECKLIST
Internet and Outlook connectivity verified
Send/Receive settings configured
Outlook cache cleared
Send/Receive groups reset
Outlook updated
Mailbox quota checked
Works in Safe Mode
Profile repaired
OST file repaired (if needed)
Issue resolved and confirmed
USEFUL COMMANDS / PATHS
Command / Path | Purpose
outlook.exe /safe | Run Outlook in Safe Mode
scanpst.exe | Scan and repair OST file
%localappdata%\Microsoft\Outlook | Outlook cache location
Win + R | Open Run dialog
Send/Receive All Folders (F9) | Send/receive emails
COMMON CAUSES AND FIXES
Cause | Fix
Connectivity issue | Check internet / account settings
Send/Receive misconfigured | Reset Send/Receive groups
Outdated Outlook | Update Outlook
Cache corruption | Clear Outlook cache
Corrupt OST file | Scan and repair OST file
Profile issue | Repair Outlook profile or recreate
KEY POINTS
Always check connectivity first.
Reset Send/Receive groups if emails are stuck.
Clear cache to fix sync and search issues.
Use Safe Mode to identify add-in or profile issues.
Repair OST file if folders are not syncing.
Verify from another client before closing the ticket.

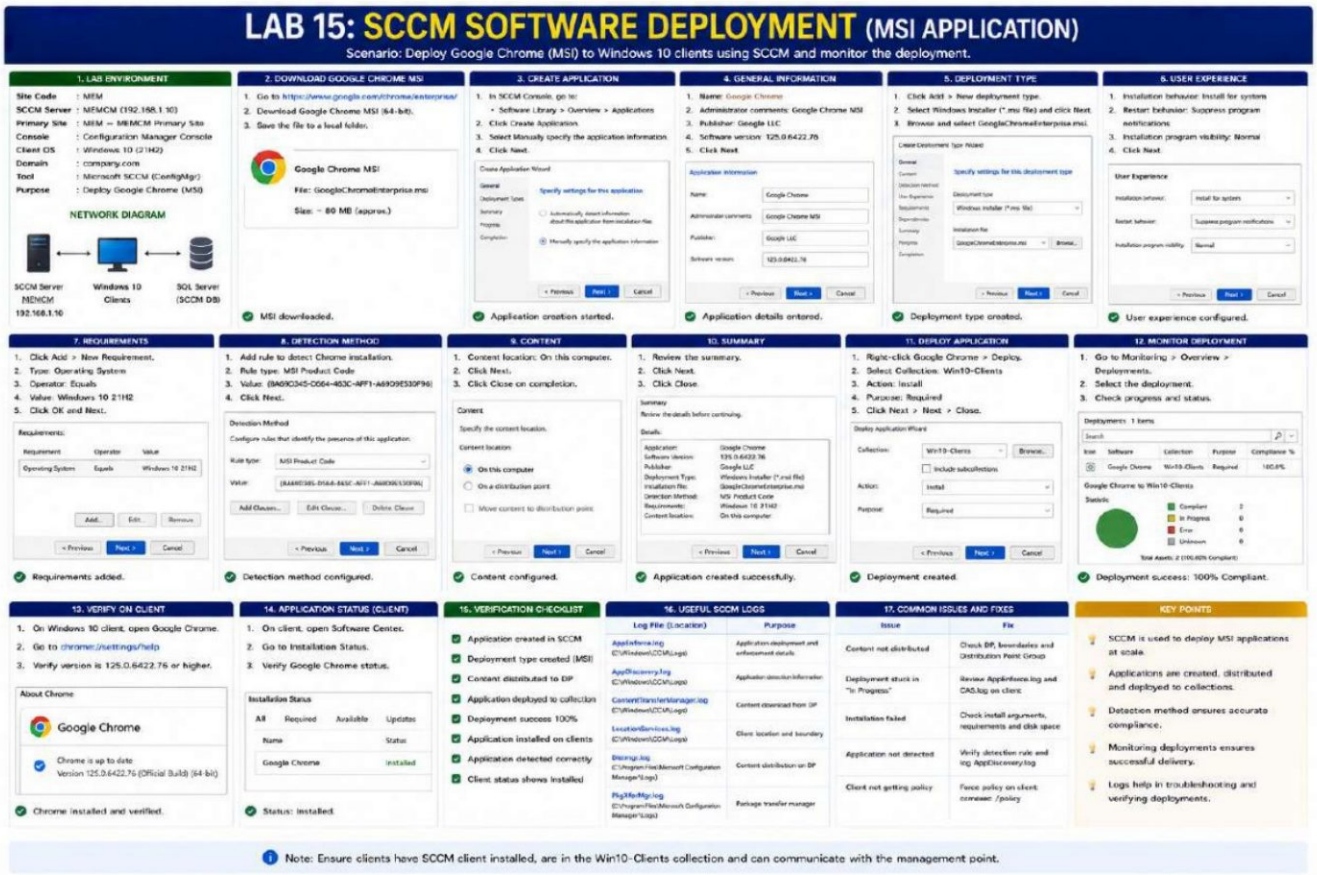

LAB 10: ONEDRIVE AND SHAREPOINT – COLLABORATION AND PERMISSIONS
Scenario: Configure OneDrive and SharePoint, create sites and libraries, sync files, share with users and manage permissions.
1. LAB ENVIRONMENT
Tenant : company.onmicrosoft.com
Admin Account : admin@company.onmicrosoft.com
User 1 : john.finance@company.com
User 2 : jane.accounts@company.com
Platform : Microsoft 365
Client PC : WIN10 (192.168.1.50)
Tool : OneDrive, SharePoint Admin Center, SharePoint, Teams (optional)
URL : https://admin.microsoft.com
NETWORK DIAGRAM
Microsoft 365
WIN10 Client 192.168.1.50
SharePoint OneDrive
2. CREATE ONEDRIVE
1. Sign in to https://portal.office.com
2. Click the app launcher (9 dots)
3. Click OneDrive.
4. Click New > Folder and name it "Project Files".
5. Upload a file and verify.
OneDrive created and file uploaded.
3. CREATE SHAREPOINT SITE
1. Go to https://admin.microsoft.com
2. Admin centers > SharePoint
3. Click Active sites > Create site
4. Choose "Team site".
5. Site name: Finance Team Site
6. Add members: john.finance@company.com, jane.accounts@company.com
7. Click Finish.
Active sites
SharePoint site created.
4. CREATE DOCUMENT LIBRARY
1. Open the site in SharePoint.
2. Click Settings > Site contents.
3. Click New > Document library.
4. Name: "Finance Documents".
5. Click Create.
Site contents > New > Document library
Document library created.
5. SYNC DOCUMENT LIBRARY
1. In the library, click Sync.
2. OneDrive will open and sync the library.
3. Files will be available offline.
4. Add or modify a file and verify sync.
Sync your files to this PC
Library synced with OneDrive.
6. SHARE FILE / FOLDER
1. In the library, select a file or folder.
2. Click Share.
3. Enter user: jane.accounts@company.com
4. Choose permission: Can edit.
5. Add a message and click Send.
Share "Budget.xlsx"
File shared successfully.
7. MANAGE PERMISSIONS
1. Click ⋯ > Manage access.
2. View existing permissions.
3. Click Advanced > Stop inheriting permissions.
4. Remove unnecessary permissions if any.
5. Click Save.
Manage access
Permissions updated.
8. CREATE SHAREPOINT LIST
1. On the site, click New > List.
2. Name: "Project Tracker".
3. Choose Blank list.
4. Add columns: Task (Single line), Owner (Person), Status (Choice), Due Date (Date).
5. Add sample items.
Project Tracker
List created and items added.
9. AUTOMATE WITH POWER AUTOMATE (OPTIONAL)
1. Go to Power Automate (make.powerautomate.com)
2. Create > Automated cloud flow.
3. Trigger: When an item is created (SharePoint).
4. Action: Send an email (V2).
5. Save and test the flow.
When an item is created
Send an email (V2)
Flow created and tested.
10. TEST COLLABORATION
1. Login as User 2 (jane.accounts@company.com).
2. Open the shared file/folder.
3. Edit and save the file.
4. Verify changes reflect for User 1.
Finance Documents > Budgets
Collaboration verified.
VERIFICATION CHECKLIST
OneDrive folder created
Files uploaded and accessible
SharePoint site created
Document library created
Library synced with OneDrive
File/folder shared with a user
Permissions configured correctly
SharePoint list created
Collaboration tested successfully
USEFUL COMMANDS / URLS
Command / URL | Purpose
https://portal.office.com | Access Microsoft 365 apps
https://admin.microsoft.com | Access Microsoft 365 admin center
https://<tenant>.sharepoint.com | Access SharePoint sites
PowerShell: Connect-SPOService -Url https://<tenant>-admin.sharepoint.com | Connect to SharePoint Online
OneDrive Sync Client | Sync libraries to local PC
COMMON CAUSES AND FIXES
Issue | Fix
Cannot find shared item | Check sharing link and permissions
Sync not working | Check OneDrive status and sign-in
Access denied | Verify user permissions
Changes not visible | Refresh page / wait and retry
Library not syncing | Check folder path and quota
List not editable | Check list permissions
KEY POINTS
Use OneDrive for personal files and SharePoint for team files.
Always share with the least privilege (Can view / Can edit).
Sync libraries to work offline and improve productivity.
Manage permissions carefully to protect data.
Lists help track and manage team information.
Automate notifications and tasks with Power Automate.

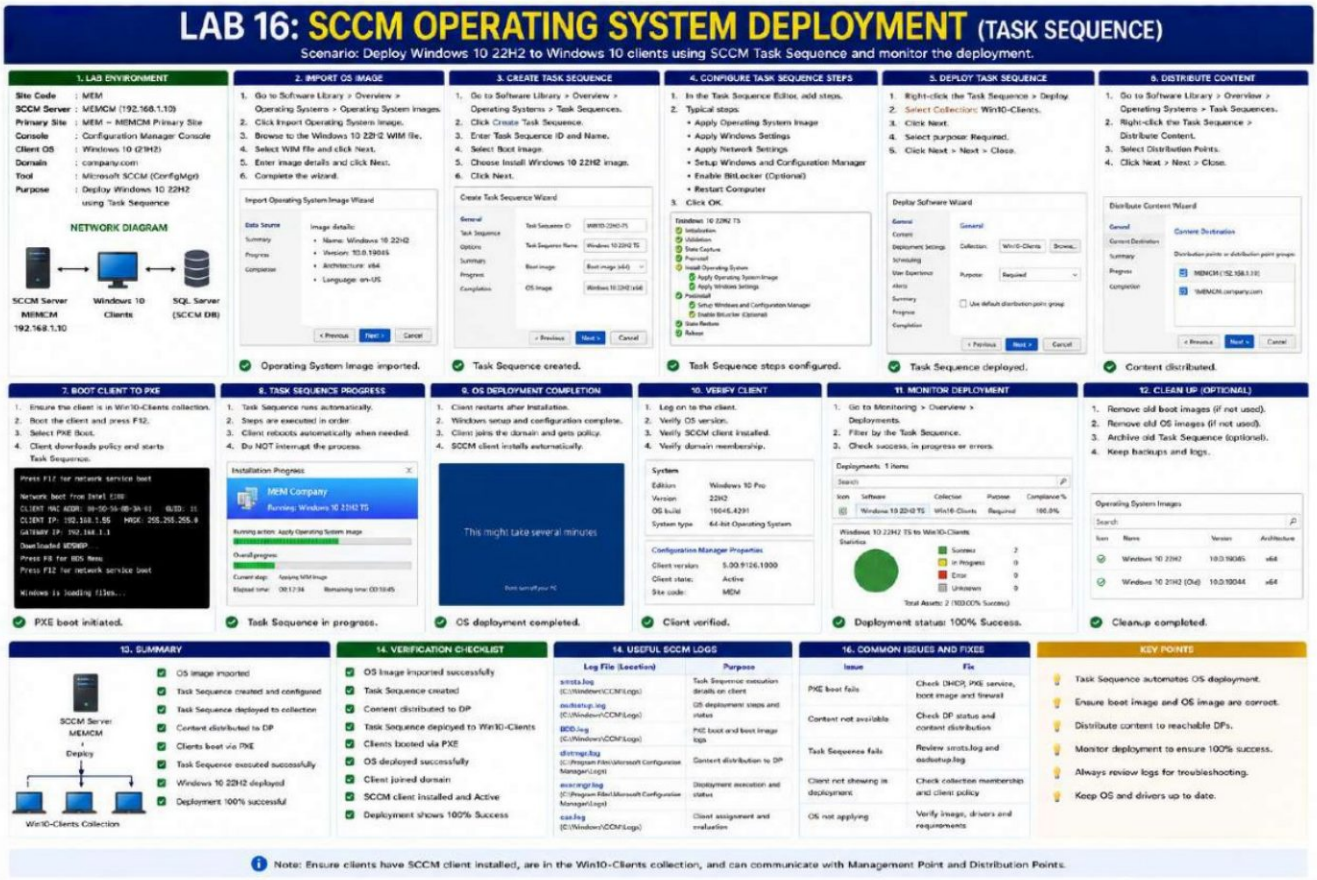

LAB 11: MICROSOFT TEAMS AND PLANNER – TEAM COLLABORATION AND TASK MANAGEMENT
Scenario: Create a Team, add members, have a channel conversation, schedule a meeting and create a plan in Planner, assign tasks and track progress.
1. LAB ENVIRONMENT
NETWORK DIAGRAM
Microsoft 365
WIN10 Client
192.168.1.50
Teams & Planner
Microsoft 365
2. CREATE A TEAM
Team "Finance Team" created.
3. ADD MEMBERS TO TEAM
Members added successfully.
4. CREATE A CHANNEL & START CONVERSATION
Conversation started in General channel.
5. SCHEDULE A MEETING
Meeting scheduled.
6. ADD TABS (FILES & PLANNER)
Files and Planner tabs added.
7. CREATE A PLAN IN PLANNER
Plan "Q2 Budget Plan" created.
8. ADD BUCKETS
Buckets added.
9. CREATE AND ASSIGN TASKS
Tasks created and assigned.
10. UPDATE TASK PROGRESS
Progress updated and tasks moved.
11. VIEW PLAN PROGRESS
Plan progress reviewed.
12. FILE SHARING IN TEAM
File uploaded and shared.
13. CHAT IN TEAM
Chat completed.
14. VERIFY ON MOBILE (OPTIONAL)
Team visible on mobile.
15. CLEAN UP (OPTIONAL)
Cleanup completed.
VERIFICATION CHECKLIST
USEFUL COMMANDS / URLS
COMMON CAUSES AND FIXES
KEY POINTS

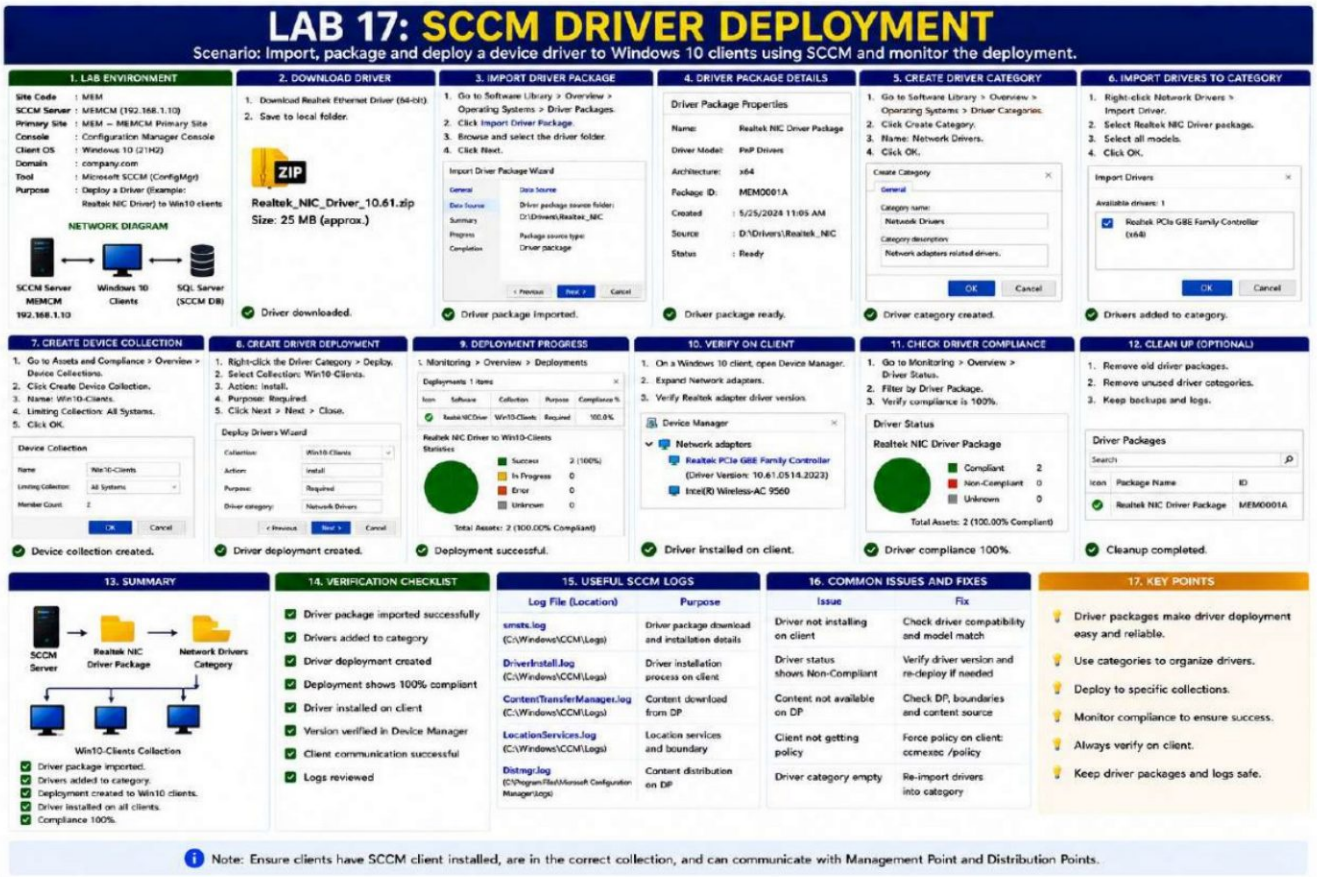

LAB 12: POWER AUTOMATE – AUTOMATE TASKS AND APPROVALS
Scenario: Create flows in Power Automate to automate tasks, send approvals and notifications.
1. LAB ENVIRONMENT
NETWORK DIAGRAM
2. SIGN IN TO POWER AUTOMATE
3. CREATE AUTOMATED CLOUD FLOW
4. CONFIGURE FLOW – NEW FILE NOTIFICATION
5. TEST NEW FILE NOTIFICATION FLOW
6. CREATE APPROVAL FLOW
7. CONFIGURE APPROVAL FLOW
8. TEST APPROVAL FLOW
9. CREATE SCHEDULED FLOW
10. CONFIGURE AND TEST SCHEDULED FLOW
11. MANAGE FLOWS
12. SOLUTIONS (OPTIONAL)
VERIFICATION CHECKLIST
USEFUL COMMANDS / URLS
COMMON CAUSES AND FIXES
KEY POINTS

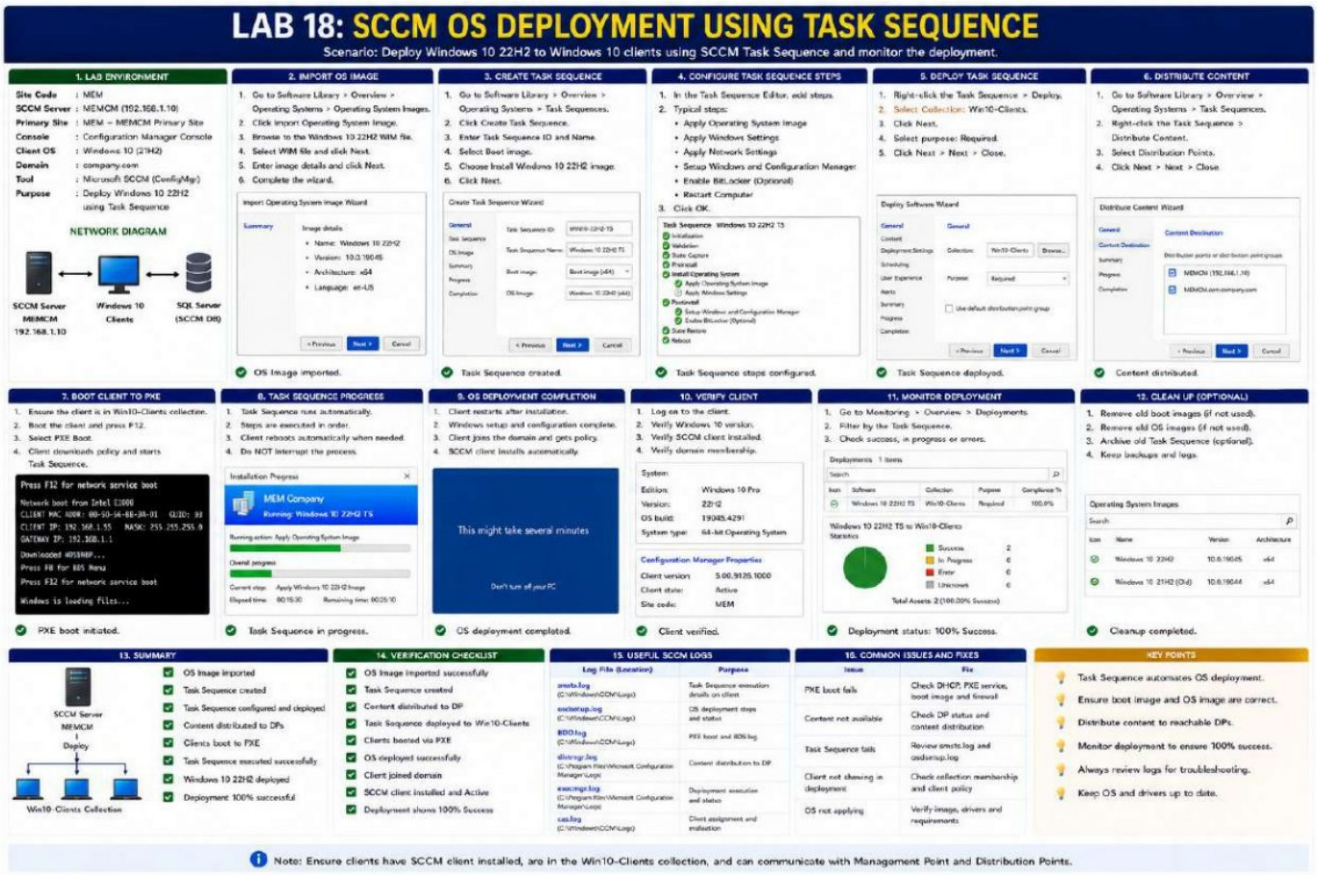

LAB 13: MICROSOFT PURVIEW DLP – CREATE POLICIES AND TEST
Scenario: Create a DLP policy to protect sensitive information, apply it to locations, test the policy and review alerts.
1. LAB ENVIRONMENT
NETWORK DIAGRAM
2. SIGN IN TO MICROSOFT PURVIEW
3. CREATE DLP POLICY
4. NAME AND ASSIGN POLICY
5. DEFINE POLICY RULE
6. SET POLICY MODE
2. REVIEW AND CREATE POLICY
8. APPLY POLICY TO LOCATIONS
9. TEST POLICY – SHAREPOINT
10. TEST POLICY – ONEDRIVE
11. TEST POLICY – EXCHANGE EMAIL
12. CHECK ALERTS
13. VIEW ALERT DETAILS
14. TURN POLICY TO ENFORCE (OPTIONAL)
15. SUMMARY
VERIFICATION CHECKLIST
USEFUL COMMANDS / URLS
COMMON CAUSES AND FIXES
KEY POINTS
Note: DLP scans run periodically. It may take 5–15 minutes (or more) for alerts to appear.

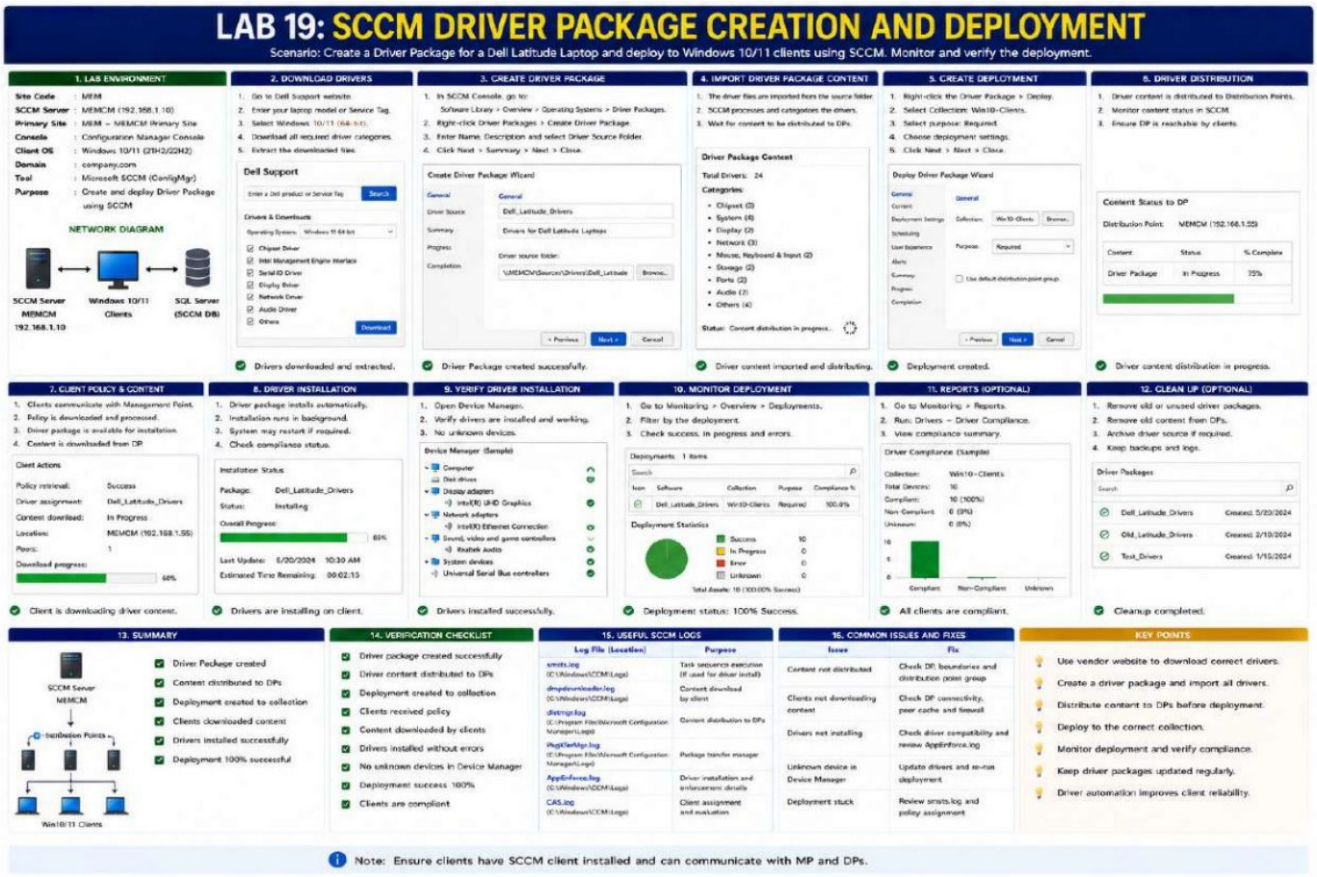

LAB 14: SCCM SOFTWARE DEPLOYMENT
Scenario: Package and deploy 7-Zip to Windows 10 clients using SCCM and monitor the deployment.
1. LAB ENVIRONMENT
Site Code : MEM
SCCM Server : MEMCM (192.168.1.10)
Primary Site : MEM – MEMCM Primary Site
Console : Configuration Manager Console
Client OS : Windows 10
Domain : company.com
Tool : Microsoft SCCM (ConfigMgr)
Purpose : Deploy 7-Zip to Windows 10 clients
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10 Clients
SQL Server (SCCM DB)
2. DOWNLOAD 7-ZIP INSTALLER
1. Go to https://www.7-zip.org/.
2. Download 7-Zip (x64) installer.
3. Save the file to a local folder.
7ZIP
7-Zip is a file archiver with a high compression ratio.
Download 7-Zip 23.01 (2023-06-20) for Windows
Link Type Windows Size
Download .exe 64-bit x64 1.5 MB
Download .exe 32-bit x86 1.1 MB
License
Installer downloaded.
3. CREATE APPLICATION
1. In SCCM Console, go to:
• Software Library > Overview > Application
2. Click Create Application.
3. Select Manually specify the application information.
4. Click Next.
Application creation started.
4. CONFIGURE APPLICATION DETAILS
1. Name: 7-Zip
2. Administrator comments: 7-Zip File Archiver
3. Publisher: Igor Pavlov
4. Software version: 23.01
5. Click Next.
Application details configured.
5. ADD DEPLOYMENT TYPE
1. Click Add > New deployment type.
2. Select Windows Installer (*.msi file) and click Next.
3. Browse and select the 7z2301-x64.msi file.
4. Click Next.
Deployment type added.
6. CONFIGURE DEPLOYMENT TYPE
1. User experience: Install for system
2. Installation behavior: Install for system
3. Restart behavior: Suppress program notifications
4. Click Next then Close.
Deployment type configured.
7. DISTRIBUTE CONTENT
1. In the application, go to the Content tab.
2. Click Distribute Content.
3. Select Distribution Point(s).
4. Click Next > Next > Close.
Content distributed to DP.
8. CREATE DEVICE COLLECTION
1. Go to Assets and Compliance > Overview > Device Collections.
2. Click Create Device Collection.
3. Name: Win10-Clients
4. Add members (or use query membership).
5. Click OK.
Device Collection
Name: Win10-Clients
Limiting Collection: All Systems
Member Count: 2
Device collection created.
9. DEPLOY APPLICATION
1. Right-click 7-Zip application > Deploy.
2. Select collection: Win10-Clients
3. Action: Install
4. Purpose: Required
5. Click Next > Next > Close.
Deployment created.
10. MONITOR DEPLOYMENT
1. Go to Monitoring > Overview > Deployments.
2. Select the 7-Zip deployment.
3. Review the overall compliance.
7-Zip Deployment to Win10-Clients
Compliant
In Progress
Error
Unknown
Deployment success: 100% Compliant
11. VERIFY ON CLIENT
1. On Windows 10 client, open Command Prompt.
2. Run: 7z
3. Verify 7-Zip is installed.
Command Prompt
C:\>7z
7-Zip 23.01 (x64)
Copyright (c) 1999-2021 Igor Pavlov
Usage: 7z
command [switches...] archive_name [-x[r[-|0]]] [+[r[-|0]]] [-o{Output_dir}] [-p{Password}] [-w[{Working_dir}]] [-bs{Msg_Id}:{Value}] [-bt] [-bb[0-3]] [-bd] [-sa{a|s|e}] [-ssc[-]] [-sdel] [-seo] [-sfx[{name}]] [-si[{name}]] [-slp{N}] [-slt] [-snh] [-snl] [-snr[{name}]] [-snt{name}] [-sns[-]] [-sse{N}] [-ssw] [-stl] [-stm{HexMask}] [-stx{Type}] [-sv{Time}] [-sw{N}] [-t{Type}] [-u[-][p#][q#][r#][x#][y#][z{FILE}]] [-v{Size}[k|m|g|t|p|e|z|y]] [-x[r[-|0]]] [-y] [-z{FILE}] [-ai[r[-|0]]{ArcPaths}] [-ax[r[-|0]]{ArcPaths}] [-bc[{Method}]] [-bh[a|c|m|p|s|t|w|x|y|z|{Method}]] [-bp{N}[b|k|m|g]] [-bsp{N}] [-bso{N}] [-bsp1{N}] [-bsp2{N}] [-ci[r[-|0]]{ExcludeArcPaths}] [-cx[r[-|0]]{ExcludeArcPaths}] [-ea{AttributeDefinition}] [-eb{BaseDir}] [-ed] [-ef{Method}] [-eh{FileHashes}] [-ei[r[-|0]]{ExcludeItems}] [-el] [-em{Method}] [-en{FileNamePrefix}] [-eo{ExcludeDirs}] [-ep[0|1|2]] [-er] [-es{Type}] [-ese{FileNameExtension}] [-ev{VolumeIndexes}] [-ev-{VolumeIndexes}] [-ev0{VolumeIndexes}] [-ev1{VolumeIndexes}] [-ev2{VolumeIndexes}] [-ev3{VolumeIndexes}] [-evb{VolumeBootMode}] [-evc{VolumeCodePage}] [-evd{VolumeDriveLetter}] [-evf{VolumeFileSystem}] [-evh{VolumeHexOffset}] [-evi{VolumeIdentifier}] [-evl{VolumeLabel}] [-evm{VolumeMediaType}] [-evn{VolumeName}] [-evp{VolumePath}] [-evs{VolumeSerialNumber}] [-evt{VolumeType}] [-ew{ExcludeWildcards}] [-ex[r[-|0]]{ExcludeArcPaths}] [-exl] [-exr{Rule}] [-ff[{FileFormats}]] [-fh{Filter}] [-fm{Filter}] [-fp{Filter}] [-fs{Filter}] [-ft{Filter}] [-fv{Filter}] [-fx{Filter}] [-gb{Bytes}] [-gc{Bytes}] [-gt{Time}] [-hb{Mode}] [-hc{Mode}] [-hd{Mode}] [-he{Mode}] [-hf{Mode}] [-hf1{Mode}] [-hf2{Mode}] [-hf3{Mode}] [-hfs{Mode}] [-hh{Mode}] [-hm{Mode}] [-hr{Mode}] [-hs{Mode}] [-ht{Mode}] [-hv{Mode}] [-hx{Mode}] [-i[r[-|0]]{IncludeItems}] [-ic{FileCodePage}] [-ie{Number}] [-im] [-ir[r[-|0]]{RecurseWildcards}] [-ir!{RecurseWildcards}] [-iw{IncludeWildcards}] [-kb] [-kc{Mode}] [-kd{Mode}] [-ke{Mode}] [-kf{Mode}] [-kh{Mode}] [-kl] [-km{Mode}] [-kn] [-ko{Mode}] [-kp{Mode}] [-kr] [-ks{Mode}] [-kt{Mode}] [-ku{Mode}] [-kv{Mode}] [-kx{Mode}] [-l] [-lc{N}] [-lp{N}] [-lr{N}] [-lt{Type}] [-m{Parameters}] [-mb{N}] [-mc{N}] [-mdp{N}] [-mf[{Method}]] [-mf-] [-mh[{Method}]] [-mh-] [-mm[{Method}]] [-mm-] [-mmt[{N}]] [-mmt-] [-mp{N}] [-mp1{N}] [-mp2{N}] [-mp3{N}] [-mp4{N}] [-mp5{N}] [-mp6{N}] [-mp7{N}] [-mp8{N}] [-mp9{N}] [-mp10{N}] [-mp11{N}] [-mp12{N}] [-mp13{N}] [-mp14{N}] [-mp15{N}] [-mp16{N}] [-mp17{N}] [-mp18{N}] [-mp19{N}] [-mp20{N}] [-mp21{N}] [-mp22{N}] [-mp23{N}] [-mp24{N}] [-mp25{N}] [-mp26{N}] [-mp27{N}] [-mp28{N}] [-mp29{N}] [-mp30{N}] [-mp31{N}] [-mp32{N}] [-mp33{N}] [-mp34{N}] [-mp35{N}] [-mp36{N}] [-mp37{N}] [-mp38{N}] [-mp39{N}] [-mp40{N}]

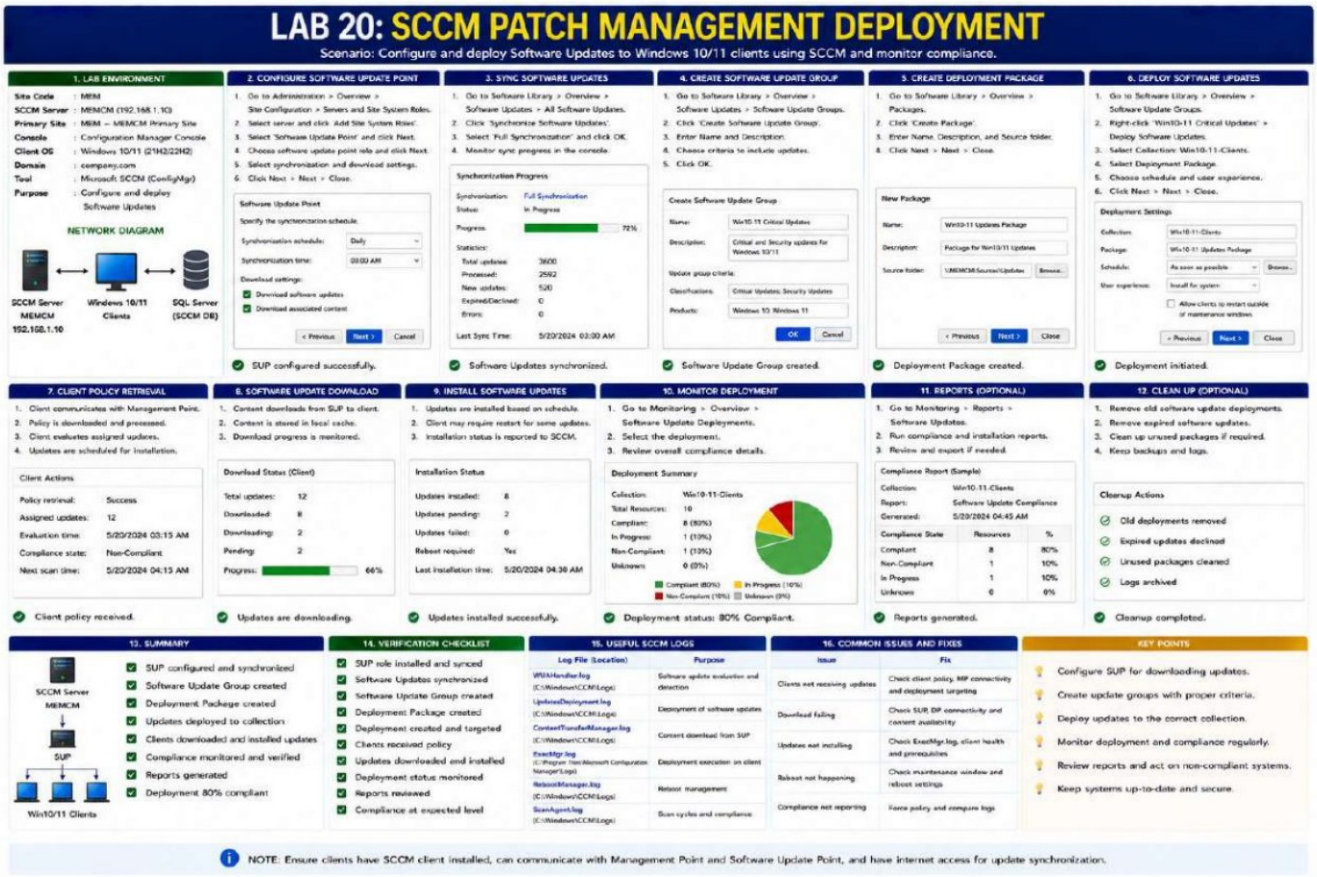

LAB 15: SCCM SOFTWARE DEPLOYMENT (MSI APPLICATION)
Scenario: Deploy Google Chrome (MSI) to Windows 10 clients using SCCM and monitor the deployment.
1. LAB ENVIRONMENT
Site Code : MEM
SCCM Server : MEMCM (192.168.1.10)
Primary Site : MEM — MEMCM Primary Site
Console : Configuration Manager Console
Client OS : Windows 10 (21H2)
Domain : company.com
Tool : Microsoft SCCM (ConfigMgr)
Purpose : Deploy Google Chrome (MSI)
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10 Clients
SQL Server (SCCM DB)
2. DOWNLOAD GOOGLE CHROME MSI
1. Go to https://www.google.com/chrome/enterprise/
2. Download Google Chrome MSI (64-bit).
3. Save the file to a local folder.
Google Chrome MSI
File: GoogleChromeEnterprise.msi
Size: ~ 80 MB (approx.)
MSI downloaded.
3. CREATE APPLICATION
1. In SCCM Console, go to:
• Software Library > Overview > Applications
2. Click Create Application.
3. Select Manually specify the application information.
4. Click Next.
Application creation started.
4. GENERAL INFORMATION
1. Name: Google Chrome
2. Administrator comments: Google Chrome MSI
3. Publisher: Google LLC
4. Software version: 125.0.6422.76
5. Click Next.
Application details entered.
5. DEPLOYMENT TYPE
1. Click Add > New deployment type.
2. Select Windows Installer (*.msi file) and click Next.
3. Browse and select GoogleChromeEnterprise.msi.
Deployment type created.
6. USER EXPERIENCE
1. Installation behavior: Install for system
2. Restart behavior: Suppress program notifications
3. Installation program visibility: Normal
4. Click Next.
User experience configured.
7. REQUIREMENTS
1. Click Add > New Requirement.
2. Type: Operating System
3. Operator: Equals
4. Value: Windows 10 21H2
5. Click OK and Next.
Requirements added.
8. DETECTION METHOD
1. Add rule to detect Chrome installation.
2. Rule type: MSI Product Code
3. Value: {8A69D345-D564-463C-AFF1-A69D9E530F96}
4. Click Next.
Detection method configured.
9. CONTENT
1. Content location: On this computer.
2. Click Next.
3. Click Close on completion.
Content configured.
10. SUMMARY
1. Review the summary.
2. Click Next.
3. Click Close.
Application created successfully.
11. DEPLOY APPLICATION
1. Right-click Google Chrome > Deploy.
2. Select Collection: Win10-Clients
3. Action: Install
4. Purpose: Required
5. Click Next > Next > Close.
Deployment created.
12. MONITOR DEPLOYMENT
1. Go to Monitoring > Overview > Deployments.
2. Select the deployment.
3. Check progress and status.
Deployment success: 100% Compliant.
13. VERIFY ON CLIENT
1. On Windows 10 client, open Google Chrome.
2. Go to chrome://settings/help
3. Verify version is 125.0.6422.76 or higher.
Chrome installed and verified.
14. APPLICATION STATUS (CLIENT)
1. On client, open Software Center.
2. Go to Installation Status.
3. Verify Google Chrome status.
Status: Installed
15. VERIFICATION CHECKLIST
Application created in SCCM
Deployment type created (MSI)
Content distributed to DP
Application deployed to collection
Deployment success 100%
Application installed on clients
Application detected correctly
Client status shows Installed
16. USEFUL SCCM LOGS
Log File (Location) | Purpose
AppEnforce.log | Application deployment and enforcement details
AppDiscovery.log | Application detection information
ContentTransferManager.log | Content download from DP
LocationServices.log | Client location and boundary
Distmgr.log | Content distribution on DP
PkgXferMgr.log | Package transfer manager
17. COMMON ISSUES AND FIXES
Issue | Fix
Content not distributed | Check DP, boundaries and Distribution Point Group
Deployment stuck in "In Progress" | Review AppEnforce.log and CAS.log on client
Installation failed | Check install arguments, requirements and disk space
Application not detected | Verify detection rule and log AppDiscovery.log
Client not getting policy | Force policy on client: ccmexec /policy
KEY POINTS
SCCM is used to deploy MSI applications at scale.
Applications are created, distributed and deployed to collections.
Detection method ensures accurate compliance.
Monitoring deployments ensures successful delivery.
Logs help in troubleshooting and verifying deployments.
Note: Ensure clients have SCCM client installed, are in the Win10-Clients collection and can communicate with the management point.

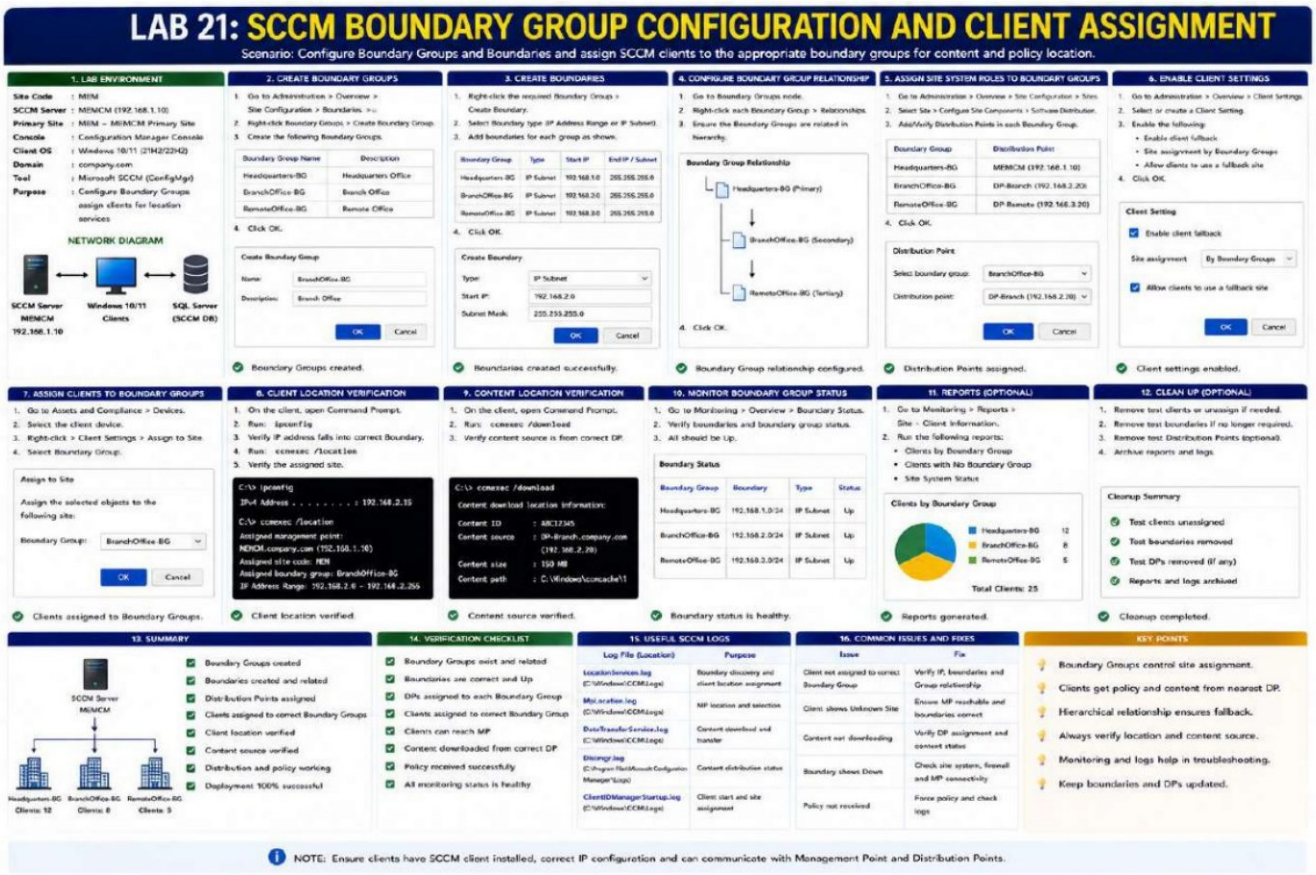

LAB 16: SCCM OPERATING SYSTEM DEPLOYMENT (TASK SEQUENCE)
Scenario: Deploy Windows 10 22H2 to Windows 10 clients using SCCM Task Sequence and monitor the deployment.
1. LAB ENVIRONMENT
2. IMPORT OS IMAGE
3. CREATE TASK SEQUENCE
4. CONFIGURE TASK SEQUENCE STEPS
5. DEPLOY TASK SEQUENCE
6. DISTRIBUTE CONTENT
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10 Clients
SQL Server (SCCM DB)
Operating System Image imported.
Task Sequence created.
Task Sequence steps configured.
Task Sequence deployed.
Content distributed.
7. BOOT CLIENT TO PXE
8. TASK SEQUENCE PROGRESS
9. OS DEPLOYMENT COMPLETION
10. VERIFY CLIENT
11. MONITOR DEPLOYMENT
12. CLEAN UP (OPTIONAL)
PXE boot initiated.
Task Sequence in progress.
OS deployment completed.
Client verified.
Deployment status: 100% Success.
Cleanup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
14. USEFUL SCCM LOGS
16. COMMON ISSUES AND FIXES
KEY POINTS
Note: Ensure clients have SCCM client installed, are in the Win10-Clients collection, and can communicate with Management Point and Distribution Points.


LAB 17: SCCM DRIVER DEPLOYMENT
Scenario: Import, package and deploy a device driver to Windows 10 clients using SCCM and monitor the deployment.
1. LAB ENVIRONMENT
Site Code : MEM
SCCM Server : MEMCM (192.168.1.10)
Primary Site : MEM – MEMCM Primary Site
Console : Configuration Manager Console
Client OS : Windows 10 (21H2)
Domain : company.com
Tool : Microsoft SCCM (ConfigMgr)
Purpose : Deploy a Driver (Example: Realtek NIC Driver) to Win10 clients
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10 Clients
SQL Server (SCCM DB)
2. DOWNLOAD DRIVER
1. Download Realtek Ethernet Driver (64-bit).
2. Save to local folder.
ZIP
Realtek_NIC_Driver_10.61.zip
Size: 25 MB (approx.)
Driver downloaded.
3. IMPORT DRIVER PACKAGE
1. Go to Software Library > Overview > Operating Systems > Driver Packages.
2. Click Import Driver Package.
3. Browse and select the driver folder.
4. Click Next.
Import Driver Package Wizard
Driver package imported.
4. DRIVER PACKAGE DETAILS
Driver Package Properties
Name: Realtek NIC Driver Package
Driver Model: PnP Drivers
Architecture: x64
Package ID: MEM0001A
Created : 5/25/2024 11:05 AM
Source : D:\Drivers\Realtek_NIC
Status : Ready
Driver package ready.
5. CREATE DRIVER CATEGORY
1. Go to Software Library > Overview > Operating Systems > Driver Categories.
2. Click Create Category.
3. Name: Network Drivers.
4. Click OK.
Create Category
Driver category created.
6. IMPORT DRIVERS TO CATEGORY
1. Right-click Network Drivers > Import Driver.
2. Select Realtek NIC Driver package.
3. Select all models.
4. Click OK.
Import Drivers
Drivers added to category.
7. CREATE DEVICE COLLECTION
1. Go to Assets and Compliance > Overview > Device Collections.
2. Click Create Device Collection.
3. Name: Win10-Clients.
4. Limiting Collection: All Systems.
5. Click OK.
Device Collection
Device collection created.
8. CREATE DRIVER DEPLOYMENT
1. Right-click the Driver Category > Deploy.
2. Select Collection: Win10-Clients.
3. Action: Install.
4. Purpose: Required.
5. Click Next > Next > Close.
Deploy Drivers Wizard
Driver deployment created.
9. DEPLOYMENT PROGRESS
Monitoring > Overview > Deployments
Deployment successful.
10. VERIFY ON CLIENT
1. On a Windows 10 client, open Device Manager.
2. Expand Network adapters.
3. Verify Realtek adapter driver version.
Device Manager
Network adapters
Realtek PCIe GBE Family Controller
(Driver Version: 10.61.0514.2023)
Intel(R) Wireless-AC 9560
Driver installed on client.
11. CHECK DRIVER COMPLIANCE
1. Go to Monitoring > Overview > Driver Status.
2. Filter by Driver Package.
3. Verify compliance is 100%.
Driver Status
Realtek NIC Driver Package
Compliant 2
Non-Compliant 0
Unknown 0
Total Assets: 2 (100.00% Compliant)
Driver compliance 100%.
12. CLEAN UP (OPTIONAL)
1. Remove old driver packages.
2. Remove unused driver categories.
3. Keep backups and logs.
Driver Packages
Cleanup completed.
13. SUMMARY
SCCM Server
Realtek NIC Driver Package
Network Drivers Category
Win10-Clients Collection
Driver package imported.
Drivers added to category.
Deployment created to Win10 clients.
Driver installed on all clients.
Compliance 100%.
14. VERIFICATION CHECKLIST
Driver package imported successfully
Drivers added to category
Driver deployment created
Deployment shows 100% compliant
Driver installed on client
Version verified in Device Manager
Client communication successful
Logs reviewed
15. USEFUL SCCM LOGS
Log File (Location) | Purpose
smsts.log (C:\Windows\CCM\Logs) | Driver package download and installation details
DriverInstall.log (C:\Windows\CCM\Logs) | Driver installation process on client
ContentTransferManager.log (C:\Windows\CCM\Logs) | Content download from DP
LocationServices.log (C:\Windows\CCM\Logs) | Location services and boundary
Distmgr.log (C:\Program Files\Microsoft Configuration Manager\Logs) | Content distribution on DP
16. COMMON ISSUES AND FIXES
Issue | Fix
Driver not installing on client | Check driver compatibility and model match
Driver status shows Non-Compliant | Verify driver version and re-deploy if needed
Content not available on DP | Check DP, boundaries and content source
Client not getting policy | Force policy on client: ccmexec /policy
Driver category empty | Re-import drivers into category
17. KEY POINTS
Driver packages make driver deployment easy and reliable.
Use categories to organize drivers.
Deploy to specific collections.
Monitor compliance to ensure success.
Always verify on client.
Keep driver packages and logs safe.
Note: Ensure clients have SCCM client installed, are in the correct collection, and can communicate with Management Point and Distribution Points.


LAB 18: SCCM OS DEPLOYMENT USING TASK SEQUENCE
Scenario: Deploy Windows 10 22H2 to Windows 10 clients using SCCM Task Sequence and monitor the deployment.
1. LAB ENVIRONMENT
2. IMPORT OS IMAGE
3. CREATE TASK SEQUENCE
4. CONFIGURE TASK SEQUENCE STEPS
5. DEPLOY TASK SEQUENCE
6. DISTRIBUTE CONTENT
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10 Clients
SQL Server (SCCM DB)
OS Image imported.
Task Sequence created.
Task Sequence steps configured.
Task Sequence deployed.
Content distributed.
7. BOOT CLIENT TO PXE
8. TASK SEQUENCE PROGRESS
9. OS DEPLOYMENT COMPLETION
10. VERIFY CLIENT
11. MONITOR DEPLOYMENT
12. CLEAN UP (OPTIONAL)
PXE boot initiated.
Task Sequence in progress.
OS deployment completed.
Client verified.
Deployment status: 100% Success.
Cleanup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
15. USEFUL SCCM LOGS
16. COMMON ISSUES AND FIXES
KEY POINTS
Note: Ensure clients have SCCM client installed, are in the Win10-Clients collection, and can communicate with Management Point and Distribution Points.


LAB 19: SCCM DRIVER PACKAGE CREATION AND DEPLOYMENT
Scenario: Create a Driver Package for a Dell Latitude Laptop and deploy to Windows 10/11 clients using SCCM. Monitor and verify the deployment.
1. LAB ENVIRONMENT
2. DOWNLOAD DRIVERS
3. CREATE DRIVER PACKAGE
4. IMPORT DRIVER PACKAGE CONTENT
5. CREATE DEPLOYMENT
6. DRIVER DISTRIBUTION
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10/11 Clients
SQL Server (SCCM DB)
Drivers downloaded and extracted.
Driver Package created successfully.
Driver content imported and distributing.
Deployment created.
Driver content distribution in progress.
7. CLIENT POLICY & CONTENT
8. DRIVER INSTALLATION
9. VERIFY DRIVER INSTALLATION
10. MONITOR DEPLOYMENT
11. REPORTS (OPTIONAL)
12. CLEAN UP (OPTIONAL)
Client is downloading driver content.
Drivers are installing on client.
Drivers installed successfully.
Deployment status: 100% Success.
All clients are compliant.
Cleanup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
15. USEFUL SCCM LOGS
16. COMMON ISSUES AND FIXES
KEY POINTS
Note: Ensure clients have SCCM client installed and can communicate with MP and DPs.


LAB 20: SCCM PATCH MANAGEMENT DEPLOYMENT
Scenario: Configure and deploy Software Updates to Windows 10/11 clients using SCCM and monitor compliance.
1. LAB ENVIRONMENT
2. CONFIGURE SOFTWARE UPDATE POINT
3. SYNC SOFTWARE UPDATES
4. CREATE SOFTWARE UPDATE GROUP
5. CREATE DEPLOYMENT PACKAGE
6. DEPLOY SOFTWARE UPDATES
NETWORK DIAGRAM
SCCM Server MEMCM 192.168.1.10
Windows 10/11 Clients
SQL Server (SCCM DB)
SUP configured successfully.
Software Updates synchronized.
Software Update Group created.
Deployment Package created.
Deployment initiated.
7. CLIENT POLICY RETRIEVAL
8. SOFTWARE UPDATE DOWNLOAD
9. INSTALL SOFTWARE UPDATES
10. MONITOR DEPLOYMENT
11. REPORTS (OPTIONAL)
12. CLEAN UP (OPTIONAL)
Client policy received.
Updates are downloading.
Updates installed successfully.
Deployment status: 80% Compliant.
Reports generated.
Cleanup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
15. USEFUL SCCM LOGS
16. COMMON ISSUES AND FIXES
KEY POINTS
NOTE: Ensure clients have SCCM client installed, can communicate with Management Point and Software Update Point, and have internet access for update synchronization.


LAB 21: SCCM BOUNDARY GROUP CONFIGURATION AND CLIENT ASSIGNMENT
Scenario: Configure Boundary Groups and Boundaries and assign SCCM clients to the appropriate boundary groups for content and policy location.
1. LAB ENVIRONMENT
2. CREATE BOUNDARY GROUPS
3. CREATE BOUNDARIES
4. CONFIGURE BOUNDARY GROUP RELATIONSHIP
5. ASSIGN SITE SYSTEM ROLES TO BOUNDARY GROUPS
6. ENABLE CLIENT SETTINGS
NETWORK DIAGRAM
Boundary Groups created.
Boundaries created successfully.
Boundary Group relationship configured.
Distribution Points assigned.
Client settings enabled.
7. ASSIGN CLIENTS TO BOUNDARY GROUPS
8. CLIENT LOCATION VERIFICATION
9. CONTENT LOCATION VERIFICATION
10. MONITOR BOUNDARY GROUP STATUS
11. REPORTS (OPTIONAL)
12. CLEAN UP (OPTIONAL)
Clients assigned to Boundary Groups.
Client location verified.
Content source verified.
Boundary status is healthy.
Reports generated.
Cleanup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
15. USEFUL SCCM LOGS
16. COMMON ISSUES AND FIXES
KEY POINTS
NOTE: Ensure clients have SCCM client installed, correct IP configuration and can communicate with Management Point and Distribution Points.

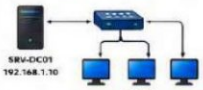
WINDOWS SERVER ADMINISTRATION – END TO END LAB

Scenario: Install, configure and manage a Windows Server 2022 system and provide core services for client computers.

1. LAB ENVIRONMENT

Server OS	: Windows Server 2022 Standard
Hostname	: SRV-DC01
IP Address	: 192.168.1.10/24
Domain	: company.com
DNS Server	: 192.168.1.10 (Local)
Gateway	: 192.168.1.1
Client OS	: Windows 10/11
Client IP Range	: 192.168.1.50 – 200

NETWORK DIAGRAM



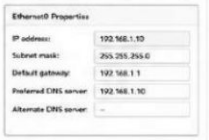
2. INSTALL WINDOWS SERVER

1. Boot from Windows Server 2022 ISO.
2. Select language, time and keyboard.
3. Click Install Now.
4. Select Windows Server 2022 Standard.
5. Accept license terms and click Next.
6. Choose Custom: Install Windows only.
7. Select disk, click Next and install.
8. After installation, set Administrator password.
9. Complete initial configuration.

Windows Server installed.

3. INITIAL CONFIGURATION

1. Rename the server to SRV-DC01.
2. Set static IP: 192.168.1.10/24, GW: 192.168.1.1.
3. Set DNS to 127.0.0.1 or 192.168.1.10.
4. Update the server.
5. Set time zone and time.
6. Enable Remote Desktop.
7. Configure Windows Firewall (Domain Profile).



Initial configuration completed.

4. PROMOTE TO DOMAIN CONTROLLER (AD DS)

1. Open Server Manager > Add Roles and Features.
2. Select Role-based installation.
3. Select this server: SRV-DC01.
4. Select Active Directory Domain Services.
5. Add required features and click Next.
6. Install and close the wizard.
7. Click the flag > Promote this server to a domain controller.
8. Select Add a new forest.
9. Enter Root domain name: company.com.
10. Set DSRM password and click Next.
11. Review and click Install.
12. Server will restart.

Server promoted to Domain Controller.

5. CONFIGURE DNS

1. DNS role is installed automatically with AD DS.
2. Verify forward lookup zone: company.com.
3. Verify reverse lookup zone: 1.168.192.in-addr.arpa.
4. Ensure DNS is listening on the server IP.

DNS configured.

6. CREATE OUS AND USERS

1. Open Active Directory Users and Computers.
2. Create the following OUs in company.com:
   - _Departments
   - _Users
   - _Groups
3. Create Users:
   - john (OU: _Users)
   - mary (OU: _Users)
4. Set passwords and uncheck "User must change password at next logon".
5. Create Groups:
   - IT-Users (Global)
   - HR-Users (Global)
6. Add John to IT-Users and mary to HR-Users.

OUs, Users and Groups created.

7. CONFIGURE DHCP SERVER

1. Add DHCP Server role.
2. Authorize DHCP in AD.
3. Create IPv4 Scope:
   - Scope Name: LAN-SCOPE
   - IP Range: 192.168.1.50 – 192.168.1.200
   - Subnet Mask: 255.255.255.0
   - Default Gateway: 192.168.1.1
   - DNS Server: 192.168.1.10
4. Activate the scope.

DHCP configured and activated.

8. CONFIGURE FILE SERVICES AND SHARES

1. Add File and Storage Services role.
2. Create a new volume (if required) or use existing.
3. Create folder: D:\Shares\CompanyData
4. Right-click > Properties > Sharing > Advanced Sharing.
5. Share name: CompanyData
6. Set NTFS Permissions:
   - IT-Users – Modify
   - HR-Users – Read
   - Administrators – Full Control
7. Verify share permissions.

File share created and permissions set.

9. GROUP POLICY BASICS

1. Open Group Policy Management.
2. Right-click company.com > Create a GPO.
3. Name: Default Domain Policy – Custom.
4. Edit the GPO and configure:
   - Password Policy
   - Desktop Background
   - Disable Control Panel
5. Link the GPO to the domain.
6. Run gpupdate /force on client.

GPO created and applied.

10. JOIN CLIENTS TO DOMAIN

1. On client, set DNS to 192.168.1.10
2. Right-click This PC > Properties > Change settings.
3. Click Change > Domain > company.com.
4. Enter domain admin credentials.
5. Restart the client.
6. Login with domain user (e.g. johni).
7. Verify access to \\SRV-DC01\CompanyData.

Client joined to domain.

11. MONITOR AND MAINTAIN

1. Check Event Viewer for errors.
2. Monitor Services and Performance.
3. Backup AD DS using Windows Server Backup.
4. Verify DHCP leases and DNS records.
5. Use Server Manager Dashboard for health.

Server is healthy and running.

12. CLEAN UP (LAB END)

1. Remove test users and groups.
2. Remove test OUs if not needed.
3. Remove file shares.
4. Delete DHCP scope (if required).
5. Remove client from domain.
6. Take final backup and logs.

Cleanup completed.

13. SUMMARY

- ☑ Windows Server installed and configured
- ☑ Active Directory Domain Services installed
- ☑ DNS configured and working
- ☑ DHCP scope created and clients get IP
- ☑ File share created with proper permissions
- ☑ Users, Groups and OUs created
- ☑ Group Policy created and applied
- ☑ Clients joined to domain and working
- ☑ Server monitored and maintained

14. VERIFICATION CHECKLIST

- ☑ Server IP and DNS working
- ☑ Can ping SRV-DC01
- ☑ Can resolve company.com
- ☑ Clients get IP from DHCP
- ☑ Can login with domain user
- ☑ Can access \\SRV-DC01\CompanyData
- ☑ GPO settings applied on client
- ☑ No critical errors in Event Viewer

15. USEFUL COMMANDS

Command	Purpose
ipconfig /all	View IP configuration
nslookup company.com	Test DNS resolution
dcdiag /v	Check Domain Controller health
gpupdate /force	Force Group Policy update
net user	List domain users
net share	List shared resources
servermanager	Open Server Manager
wbadmin start backup -backuptarget:D:	Start system backup

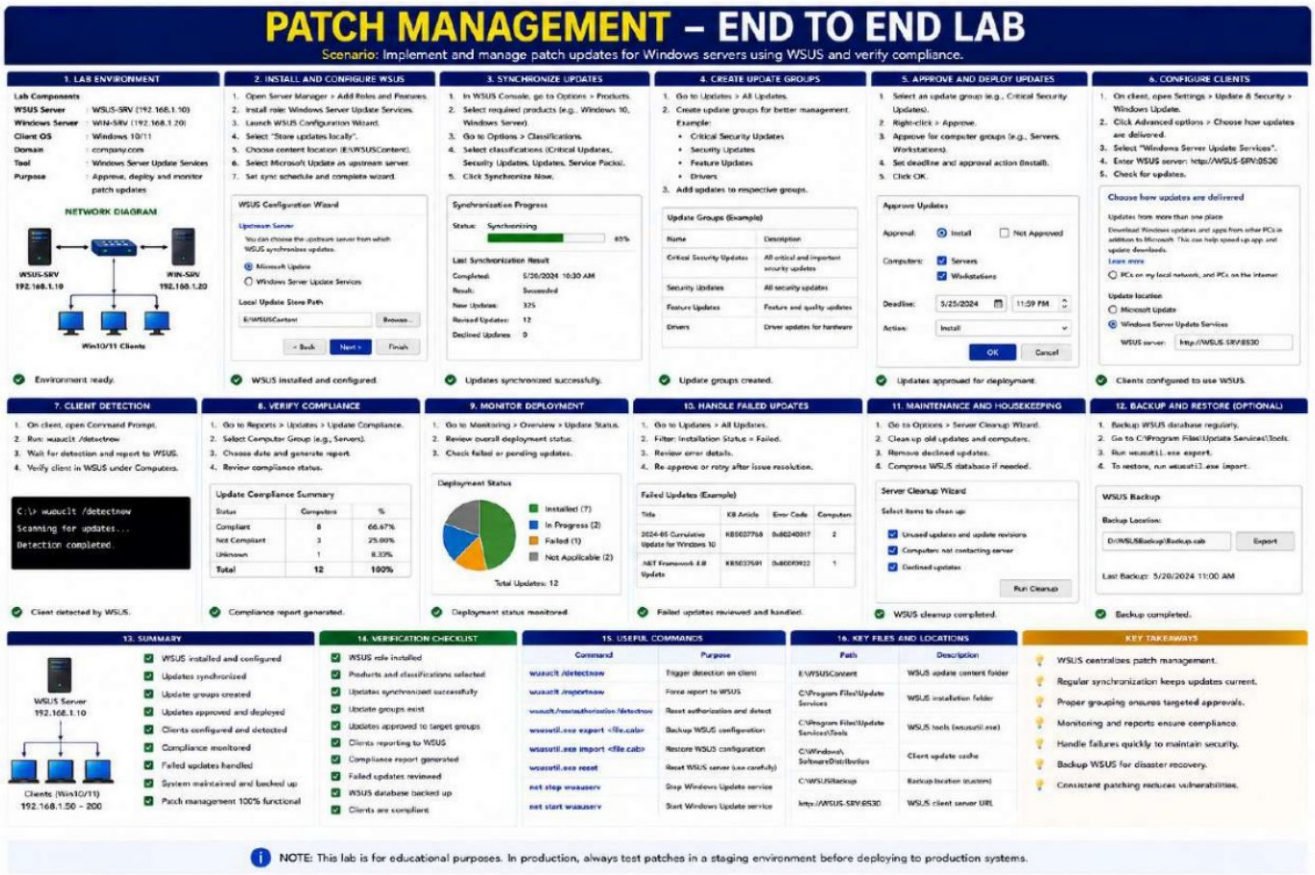
16. KEY FILES AND TOOLS

Tool / File	Purpose
Server Manager	Manage roles and features
Active Directory Users and Computers	Manage AD objects
DNS Manager	Manage DNS zones
DHCP Manager	Manage DHCP scopes
Group Policy Management	Create and manage GPOs
Event Viewer	View server logs
Windows Server Backup	Backup and restore

KEY POINTS

- AD DS is the core of Windows Server environment.
- DNS is required for domain functionality.
- DHCP automates IP address assignment.
- File sharing must have proper NTFS and Share permissions.
- Group Policy ensures consistent configuration.
- Regular monitoring and backup are essential.
- Always follow least privilege security principle.

NOTE: This lab is for educational purposes. In production, follow organizational policies, security best practices, and perform proper backups before making changes.


PATCH MANAGEMENT – END TO END LAB
Scenario: Implement and manage patch updates for Windows servers using WSUS and verify compliance.
1. LAB ENVIRONMENT
2. INSTALL AND CONFIGURE WSUS
3. SYNCHRONIZE UPDATES
4. CREATE UPDATE GROUPS
5. APPROVE AND DEPLOY UPDATES
6. CONFIGURE CLIENTS
NETWORK DIAGRAM
Environment ready.
WSUS installed and configured.
Updates synchronized successfully.
Update groups created.
Updates approved for deployment.
Clients configured to use WSUS.
7. CLIENT DETECTION
8. VERIFY COMPLIANCE
9. MONITOR DEPLOYMENT
10. HANDLE FAILED UPDATES
11. MAINTENANCE AND HOUSEKEEPING
12. BACKUP AND RESTORE (OPTIONAL)
Client detected by WSUS.
Compliance report generated.
Deployment status monitored.
Failed updates reviewed and handled.
WSUS cleanup completed.
Backup completed.
13. SUMMARY
14. VERIFICATION CHECKLIST
15. USEFUL COMMANDS
16. KEY FILES AND LOCATIONS
KEY TAKEAWAYS
NOTE: This lab is for educational purposes. In production, always test patches in a staging environment before deploying to production systems.